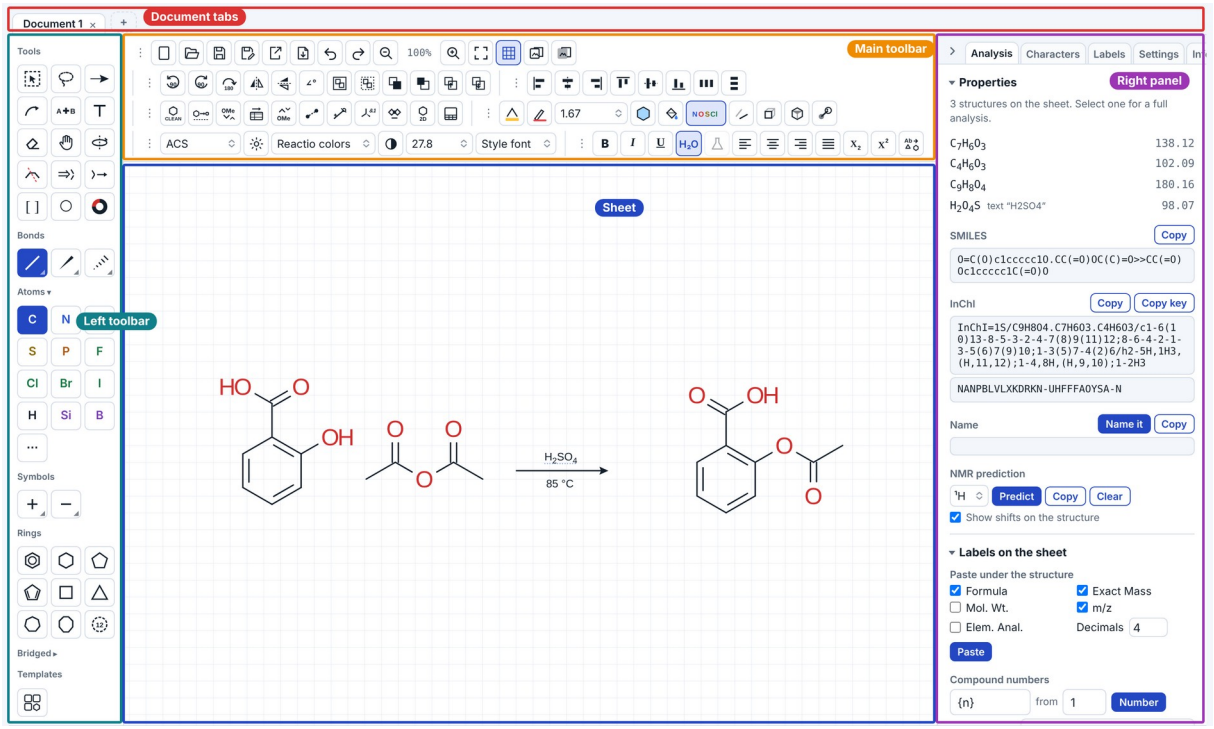
Getting started with Reactio Draw

Reactio Draw is a free editor for chemical structures and reactions, with a reagent table for every reaction. It runs as one web page or as an app for Windows, Mac and Linux.

It is **free and open source** (GNU AGPL), works **fully offline** — nothing is sent anywhere — and is **light**: the whole editor is one 4.1 MB web page (6.3 MB as a single offline file with the NMR tables), and the apps are about 20–28 MB to download.

This guide describes every button, key and function of Reactio Draw, each in a few lines. Chapter 6 is about the **API**, chapter 7 about the Reactio Draw tab in Word.

The window has four parts: the **document tabs** at the top, the **main toolbar** under them, the **left toolbar** with the drawing tools, and the **right panel** with information and settings (tabs Analysis, Characters, Labels, Settings and Info).



Installation

Windows 11

- Unzip ReactioDraw-Windows-x64.zip (ReactioDraw-Windows-ARM.zip on Windows on ARM, e.g. Parallels on a Mac) and copy the whole “Reactio Draw” folder to your own programs folder, **%LOCALAPPDATA%\Programs** (Win+R, paste it, Enter; no administrator rights needed). Word remembers where the program is, so choose the place before the first start.
- Run **ReactioDraw.exe** (ReactioDraw-ARM.exe on ARM). Keep everything in the folder together.
- If Windows shows “Windows protected your PC”, click More info → Run anyway: the program has no paid certificate. If Microsoft Defender calls it a threat, it is a false alarm on an unsigned program: allow it in Windows Security → Protection history.
- On the first start it registers itself for Word and PowerPoint, with no installer, and adds the **Reactio Draw tab** to Word (restart Word once). Reactio Draw documents (.reactio) then open on double-click. Documents and settings are in %LOCALAPPDATA%\ReactioDraw. To uninstall: “Uninstall Reactio Draw.cmd”, then delete the folder.

Mac

- Unzip ReactioDraw-Mac.zip and double-click **Install Reactio Draw.command**. If macOS says it cannot check the file: System Settings → Privacy & Security → Open Anyway (older macOS: right-click → Open).
- The installer builds the app on your Mac, puts it in Applications and opens it. If Apple’s free Command Line Tools are missing, it offers to install them; then run it again.
- When Word is on the Mac, the installer also adds the **Reactio Draw tab** to Word. **Install Word tab.command** in the folder adds it again (for example after moving the app); run it with --remove to take the tab away.

Linux

- Unpack ReactioDraw-Linux-x64.tar.gz (or -ARM), put the “Reactio Draw” folder where it should stay and run **./install.sh** in it: Reactio Draw appears in the applications menu. It needs WebKitGTK, which most desktops have; install.sh says how to get it if not.

Browser

- Unzip ReactioDraw-Web.zip and open **Reactio Draw.html** in Chrome, Edge, Safari or Firefox: nothing to install, no internet needed. Documents stay in that browser. Systematic names and the local API need the apps.

Additional software (apps)

- **Paste name** for systematic names needs **Java 8 or newer** (e.g. Eclipse Temurin from adoptium.net; Linux: sudo apt install default-jre). Common names (aspirin, THF, DIPEA…) work without it.
- **Name it** needs **Python 3.11 or newer with opencl atura**, and Java for the check that reads each name back with OPSIN. Windows: `py -m pip install openclatura`. Mac: `python3.13 -m venv ~/.reactio-python && ~/.reactio-python/bin/pip install openclatura`, then quit and reopen the app. Linux: `python3 -m venv ~/.reactio-python && ~/.reactio-python/bin/pip install openclatura`.
- Everything else — drawing, analysis, NMR prediction, 3D — needs nothing extra.

1. Document tabs

Every open document has its own tab. The usual shortcuts work:

Keys	Action
Ctrl+N	new document
Ctrl+O	open a file
Ctrl+S	save
Ctrl+Shift+S	save as
Ctrl+W	close the tab (the × next to the name does the same)
Ctrl+Shift+T	reopen a closed tab (the last 10)
Ctrl+Tab, Ctrl+Shift+Tab	next / previous tab (^Tab on a Mac too)
Alt+1 … Alt+8, Alt+9	go to that tab; Alt+9 goes to the last one (⌥ on a Mac)

On a Mac use ⌘ instead of Ctrl. **In the browser version, be careful with Ctrl+W:** the browser may close its own tab rather than the document inside Reactio Draw. The browser also keeps Ctrl+Tab (and, on some systems, Alt+1…9) for its own tabs; in the desktop apps they switch the documents.

When the tabs do not all fit, the **mouse wheel** over them scrolls the row.

A **yellow dot** next to a name means unsaved changes. Closing such a tab asks whether to save them. Unsaved work is kept in a temporary copy and only written to your file when you save.

When the program starts, only documents left with unsaved changes come back (after a crash, a power cut, or a browser tab closed anyway). Documents that were saved are not reopened; with nothing left, the program starts with one new, empty document.

2. Main toolbar

All buttons are grouped. A group can be moved, to another place or to the left toolbar, by dragging its grip ⋮; single buttons can be dragged between groups too. **Settings** → **Reset button layout** puts everything back. The layout is remembered and can be exported with your settings.

⋮ the grip of a group: drag it to move the group; on the top bar a click folds the group into one named menu, another click unfolds it.

With **Settings** → **Interface** → **Top bar** → **Compact**, the top bar keeps the file buttons and puts everything else into four menus in one row: Arrange, Structure, Style and Text.



Messages (what was done, what could not be done, warnings) appear in a small box over the bottom of the sheet for a few seconds, or in the line of the stoichiometry grid under the editor when that grid is shown.

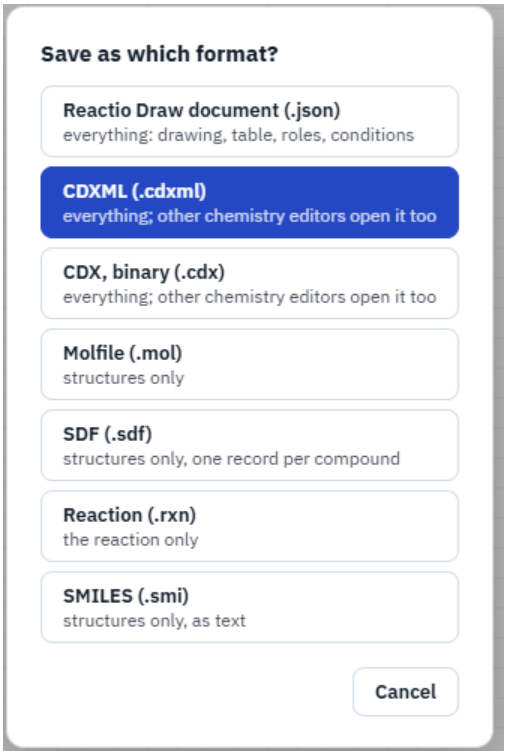
Files

 **New document** (Ctrl+N), **Open** (Ctrl+O), **Save** (Ctrl+S) and **Save as** (Ctrl+Shift+S).

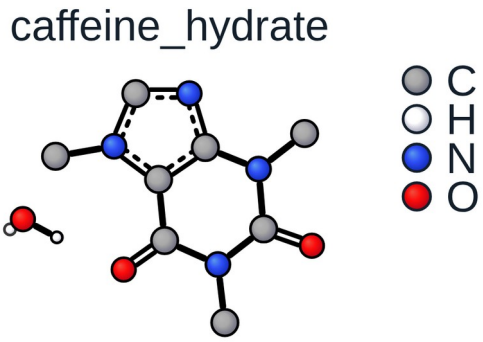
An opened file is shown whole: the view fits the drawing to the window, as **Fit to window** does.

Save as asks for the format. Reactio Draw document (.reactio), CDXML and CDX keep everything: drawing, reagent table, roles and conditions; Molfile, SDF and SMILES keep the structures only, RXN the reaction. Pictures placed on the sheet go into CDXML and CDX as pictures other editors show

(PNG, JPEG or GIF), and pictures in CDX and CDXML files (also TIFF) open in Reactio Draw. Molfiles and SD files open in both V2000 and V3000; with stereo groups on the sheet they are saved as V3000 (V2000 cannot hold the groups), and SMILES as CXSMILES.



Crystal structures (CIF), test version. Open and drag-and-drop also take .cif files. Every data block goes on the sheet in its measured 3D geometry, drawn ball and stick, captioned with its name. The molecules of one block (a salt with its ions, a solvent) stay where they are in the crystal and are grouped: they move and turn together. X-ray data have no bond orders: they are worked out from the hydrogens in the file and the usual valences, with charges (N+, B−, PF6−) and the charge of a metal that balances its ions, so check them. Reading CIF is a test version, not fully checked yet; the message after opening says so.

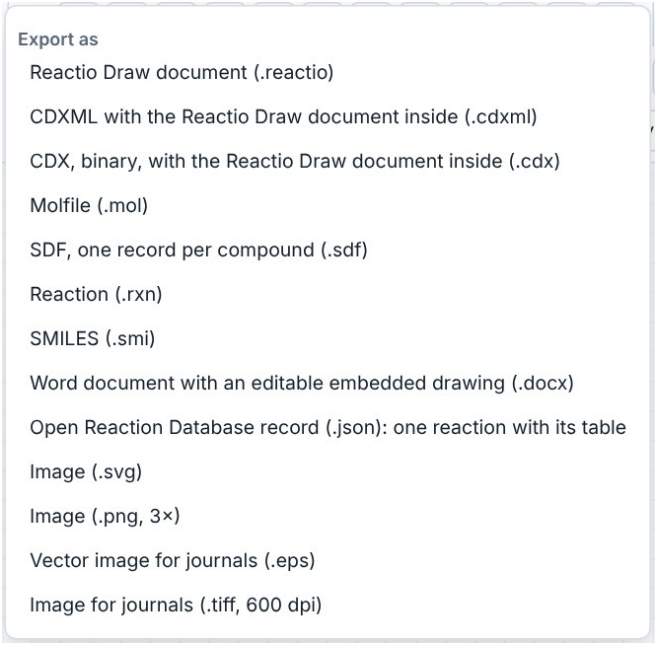


A crystal structure keeps its measured shape: moving, turning, 3D rotate, a flip and resizing move it as a whole, and an atom dragged goes back. **Clean Up** and **Clean Up 2D** work on a copy put beside it, so the flat formula with wedges comes from the crystal structure while the measured one stays. Atoms, ions and solvent can be deleted; the rest keep their places.

3D Molfiles and SD files keep their z coordinates: they open in their own 3D geometry, so 3D rotate and the 3D styles show the conformer of the file.

Export as…: Reactio Draw document (.reactio), CDXML, CDX, Molfile, SDF, RXN, SMILES, Word document with an editable drawing (.docx), Open Reaction Database record (.json), SVG, PNG, EPS and TIFF.

Unlike Save as, Export makes a separate file and the document stays tied to its own file. For example, you work in reaction.cdxml and export a PNG for a paper: Ctrl+S still saves to reaction.cdxml.



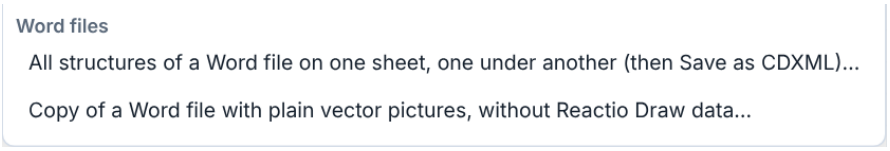
File menu: copy to the clipboard as CDXML, Molfile, RXN, SMILES, InChI and InChIKey, or as a picture for Word / PowerPoint; export and import of your workspace (hotkeys, button layout, labels and templates) as one .json file, to keep it or to give it to a colleague; open a file; insert a picture.

Pasting and dropping. Ctrl+V pastes text, SMILES, molfiles, CDXML and pictures at the place of the pointer. A picture can also be dropped onto the sheet or inserted from the File menu. A CDXML or CDX file dropped onto the sheet opens in a new tab; a Word file (.docx) opens with all its drawings on one sheet.

Reactio Draw documents are saved as **.reactio** files: the whole document, with the reagent table, roles and conditions. The Windows, Mac and Linux apps open them on double-click. Files saved as .json by earlier versions still open.

Word files

The last part of the File menu works on a saved Word document (.docx):



- **All structures of a Word file on one sheet**: every drawing of the file, in the order of the text, goes onto a new sheet one under another, without overlapping: Reactio Draw objects (Windows), Reactio Draw pictures (Mac) and objects of other chemistry editors. Save the sheet as CDXML to have them all in one file.
- **Copy of a Word file with plain vector pictures**: a copy of the document, "… (plain).docx", in which every Reactio Draw drawing is a plain vector picture with no Reactio Draw data left in it, nothing embedded and nothing in the alt text. Objects from Windows keep their own vector picture; pictures from a Mac get a vector (SVG) picture, which Word 2016 and later show. Objects of other editors stay as they are.

The Windows and Mac apps also add a Reactio Draw tab to Word, with these jobs and the editing of single drawings: see 7. Reactio Draw tab in Word.

Undo, zoom, view and copying



- **Undo** (Ctrl+Z) and **Redo** (Ctrl+Y).
- **Zoom out / zoom in.** Ctrl+mouse wheel or a pinch on a trackpad zoom too; the zoom speed is set in Settings.
- **Touch screens.** On the sheet, fingers and a pen only move it (one finger) and zoom it (two fingers); buttons, menus, keys and the API stay available. Draw with the mouse or trackpad, the keys or the API. Computers with a mouse or a trackpad work as usual.
- **Fit to window** (Ctrl+0) and **show / hide the grid** of the sheet.
- **Copy as a picture** for Word or PowerPoint (Ctrl+Shift+C), and **with the background** of the sheet (for dark or coloured paper). Pictures leaving the program never carry the on-screen warnings (red valence circles, structure-check marks, boxes of unknown labels).

Into Word and PowerPoint. Select the drawing (or nothing, for the whole sheet), press Ctrl+C and paste it into Word or PowerPoint with Ctrl+V.

- **Windows:** it arrives as a Reactio Draw object with a vector picture. Double-click it in Word to open it in Reactio Draw; your edits go back into Word when you press Ctrl+S or close the window. The object can also be copied from Word and pasted back into Reactio Draw.
- **Mac:** ⌘C → ⌘V gives a picture with the Reactio Draw drawing hidden inside it. To edit it, double-click the picture, or select it and press **Edit drawing** on the Reactio Draw tab in Word: it opens in Reactio Draw, and Save (⌘S) puts the new picture back in its place. Or copy the picture in Word (⌘C) and paste it into Reactio Draw (⌘V): structures, table and conditions come back.
- **Linux:** it is pasted as a plain picture.

Objects



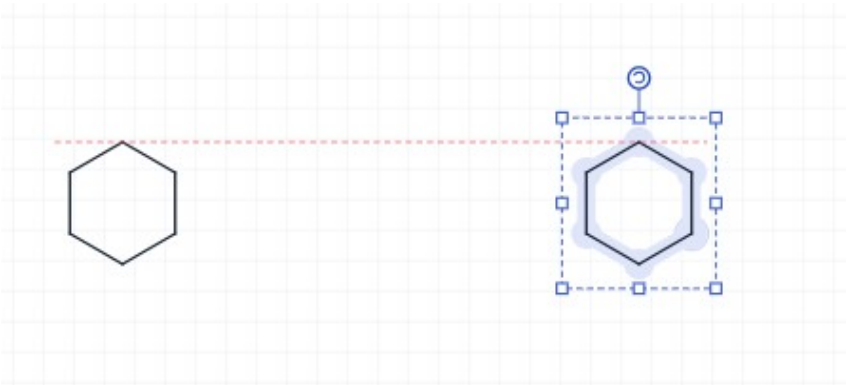
Rotate by 90°, 180° or any angle, flip horizontally or vertically, **group** and **ungroup** (Ctrl+G), and the order of overlapping objects (to front, to back, forward, backward).

A **group** counts as one object everywhere: alignment and distribution use its own edges and centre (what is inside stays as it is), and the alignment guides snap to the group as a whole.

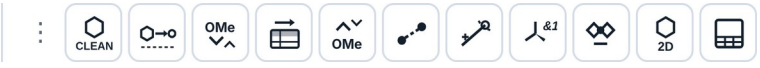
Alignment



Align the selected objects (left, centre, right, top, middle, bottom) and distribute them evenly. Texts are measured in their own font, so a centred caption or compound number sits exactly under the middle of its structure. While you drag, **smart guides** show when edges or centres line up with other objects (Alt switches them off). Selected objects can also be moved with the arrow keys.

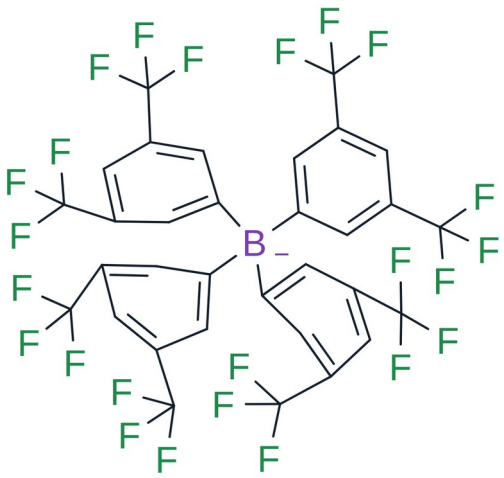


Structure

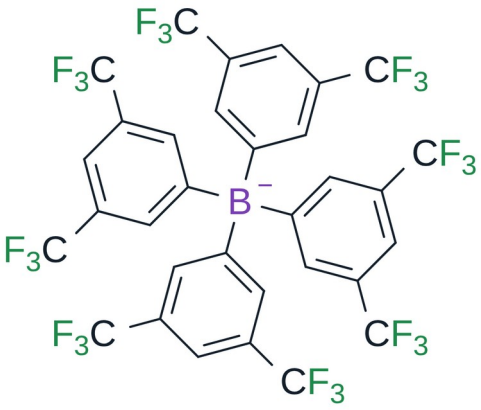


Clean Up Structure (Ctrl+Shift+K): redraws the selected structures with standard bond lengths and angles. Fused rings come out as they are usually drawn: regular hexagons on the grid, a five-membered ring between them slightly wide (acenaphthene with its naphthalene flat), symmetric when the system can be (fluorene, carbazole), with the substituents grown from the ring system. The configuration is kept: R/S at every centre with a wedge and E/Z at every double bond are checked after the new layout, and anything that turned over is put back; if that cannot be done, the structure stays as drawn and a message says so.

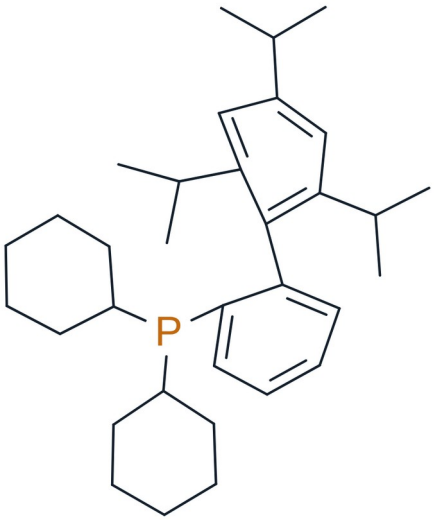
When a structure has little room, Clean Up helps itself in three ways. Three or four rings on one atom (BArF−, Ph4C, Ph3SiR) are set out as a cross, and a CF3 or tBu end turns a little out of the next group's way. Labels that cover each other are cleared by turning the whole structure slightly (at most 45°). Groups that still do not fit — their atoms land on others, their bonds cross, or their labels cover each other however the structure is turned — are written as labels: CF3, OMe, tBu, iPr, PCy2, CO2Et and the other groups of Contract Labels (not Ph), and every copy of that group in the structure goes the same way. When a ring has to be bent to make room, every group of the structure is written as its label: XPhos comes out with two regular rings, three iPr and Cy2P. Pressed again on the same structure, not changed since, Clean Up gives another layout (a branch turned over, the structure grown from another atom), each with the configuration kept; the message says which of how many it is. A structure with stereocentres or E/Z double bonds is left drawn out; a structure with room is not changed.



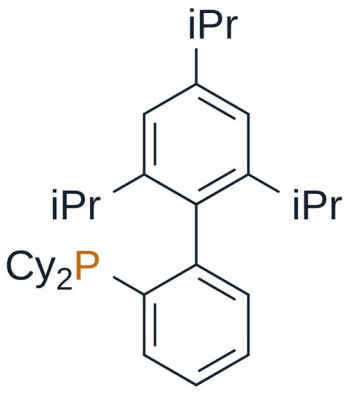
Crowded: rings and CF3 groups run into each other



After Clean Up



Crowded: a ring squashed to make room



After Clean Up

An H drawn on N, O or S goes into the label (NH, OH, SH, with no bond line); an H on carbon or on a wedge stays. A drawn methoxy (O–CH3) and a methyl drawn as a bond end can be written as labels too: **Settings** → **Drawing** (see chapter 4).

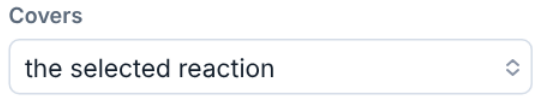
Clean Up Reaction (Ctrl+Shift+X): lines up the whole scheme — reactants, arrow, products and plus signs. Text written over the arrow (reagents) or under it (solvent, conditions) stays there, centred, and the arrow grows to fit it; reagents written as text left of the arrow line up with the reactants.

Clean Up 2D: lays the selected structures out flat. A structure turned in 3D (3D shape, 3D rotation) becomes a flat drawing again, its configuration kept as wedges and hashes. Bridged rings such as norbornane or camphor are drawn flat: a regular hexagon with the bridge inside it. Clean Up Structure keeps such systems in perspective.

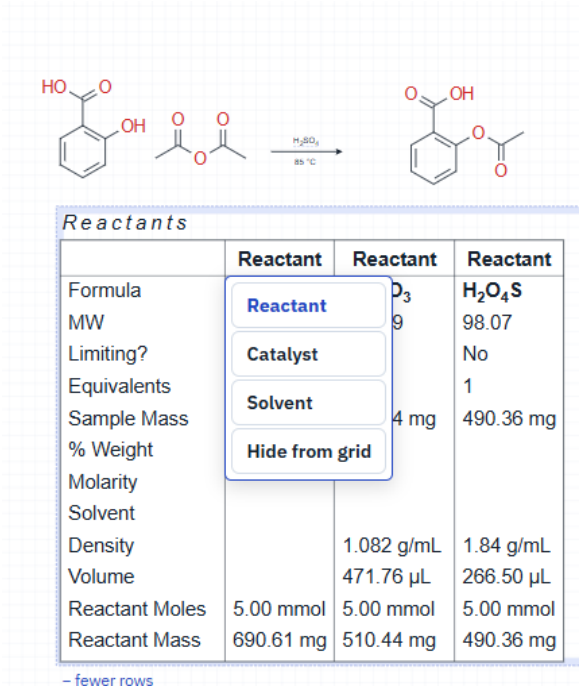
Expand Label (abbreviation → atoms, e.g. OTBS drawn out) and **Contract Labels** (known groups drawn as atoms → their label).

Reagent table on the sheet: puts the table of the reaction on the sheet (click again to remove).

Which compounds go into the table depends on Settings → Reagent table → **Covers**. **The whole sheet** suits a document with one reaction: every compound drawn on the sheet goes into the table. **The selected reaction** (the setting a new installation starts with): select the scheme first, then make the table; structures drawn later do not join it, but text written over or under its arrow does. With nothing selected, a sheet with one reaction gets its table all the same.



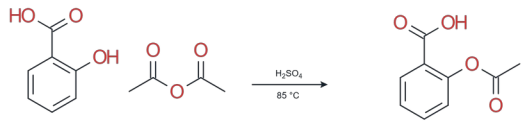
A new table is empty, apart from the density of known solvents: the amounts are yours to enter. Every compound starts as a reactant; make a solvent or a catalyst of it by clicking its cell in the first row (AcOH written under the arrow, for example, is a reactant until you choose Solvent) (Hide from grid takes a compound out of the table).



- **Scale:** enter the amount of the limiting reagent (mass, volume or moles); everything else is counted from it.
- **Solvent:** the density fills itself in; enter the volume and the concentration of the limiting reagent is calculated, or enter the concentration and get the volume.

- **Catalyst:** equivalents are in mol %.
- **Equivalents** are 1 until you enter others. **Solvent:** its equivalents (moles per mole of the limiting reagent) set its volume, from the density.
- **Reagents:** enter the equivalents and the mass is calculated, or the mass and the equivalents are. For a liquid enter its density and then either the equivalents or the volume. Without density or molarity a volume can still be entered: it is kept and follows the amounts (twice the equivalents or twice the limiting reagent, twice the volume); until you enter its equivalents, they, its moles and mass stay empty. Enter the density later and they are calculated from that volume.
- **Solutions:** enter the molarity; the Solvent row takes the solvent of the solution (not always the reaction solvent).
- **Purity** of starting materials and products, in %, is taken into the calculations.
- **After the reaction** enter the mass of the product obtained: the yield is calculated.
- **Limiting reagent:** the first reactant at first; to choose another, click “No” in the Limiting? row of its column (in the table under the editor: its round button).
- **Several steps:** with Covers = the whole sheet, a scheme with several arrows gets a table of its own for each step.

What you actually measured is shown in red. Before the reaction, mark the mass or the volume you will weigh or measure out: Alt+click that value. So it is clear that a liquid was taken by volume and not by mass.



Reactants

	Reactant	Reactant	Solvent
Formula	$C_7H_6O_3$	$C_4H_6O_3$	H_2O_4S
MW	138.12	102.09	98.07
Limiting?	***Yes***	No	—
Equivalents	1	1.959	—
Sample Mass	**690.61 mg**	**1.00 g**	46.00 g
% Weight			—
Molarity			200.00 mM
Solvent			—
Density			1.84 g/mL
Volume			**25.00 mL**
Reactant Moles	5.00 mmol	9.80 mmol	469.04 mmol
Reactant Mass	690.61 mg	1.00 g	—

Products

	Product
Formula	$C_9H_8O_4$
MW	180.16
Equivalents	1
Expected Mass	900.79 mg
Expected Moles	5.00 mmol
Measured Mass	
Purity	
Product Mass	
Product Moles	
% Yield	

Conditions

Temperature	85 °C
Time	
Atmosphere	Air
Other	

On the right are the **conditions**: temperature (rt, reflux, 0 °C, −78 °C or any value), time, atmosphere and other (light with its wavelength, microwave, ultrasound…; several can be chosen). Values can be typed as plain numbers, which take the unit already used in that row, or with a unit: 1 g, 1 mL, 10 min, 1 h.

 **Bond the two selected atoms** and clean up the result (Ctrl+Shift+L).

 **Free bonds**: bonds of any length and angle (otherwise they snap to the standard length and 15° steps). Holding Alt while drawing does the same once.



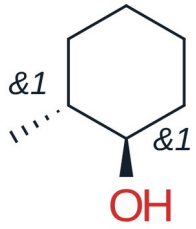
Stereo groups. The **Stereo group** button (Structure group, after Free bonds) marks the selected stereocentres: **abs** (Absolute configuration) — this enantiomer; **&1** (Racemate) — both configurations together, e.g. a racemate, the wedges give the relative configuration; **or1** (Relative configuration) — one of the two, not known which. Centres in one group change together (their configuration relative to each other is as drawn), so usually all the centres of a molecule go into one group; separate groups

(&1, &2) are independent, e.g. a mixture of diastereomers. Draw the wedge first. The label stays beside the centre and travels with CDXML/CDX, MOL/SDF (V3000) and SMILES (CXSMILES, e.g. |&1:1|).

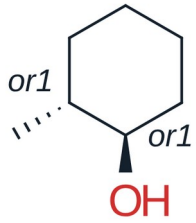
Merge structures (the last button of the Structure group): atoms of two structures drawn on top of each other become one atom, and bonds lying on top of each other become one bond (the higher order kept); a labelled or hetero atom is kept over a plain carbon. With two structures selected and nothing overlapping, the second is moved so that their two closest atoms meet, and those merge. With nothing selected it works on the whole sheet.

Enantiomers and diastereomers. With n stereocentres there are up to 2^n stereoisomers. Mirror images are enantiomers; all the other pairs are diastereomers, which differ in physical properties and NMR. The groups say which of these stereoisomers are in the flask.

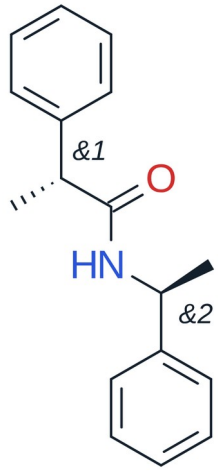
One group or several? Put centres into one group when you know how they stand to each other — they change together. Use separate groups for centres that vary independently. An enantiopure part stays **abs**. Examples:



Racemate, one group. 2-Methylcyclohexanol has two stereocentres, so four stereoisomers: the trans pair (1R,2R)/(1S,2S) and the cis pair (1R,2S)/(1S,2R). The two members of a pair are enantiomers; trans and cis are diastereomers. Both centres &1: the trans pair only, this compound and its mirror image 1:1, no cis.

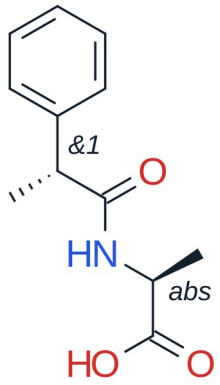


Relative configuration. Both centres or1: one of the two trans enantiomers, not known which (e.g. one peak from a chiral column); no cis, no mirror image.



Two racemic parts, two groups. Amide of racemic 2-phenylpropanoic acid and racemic 1-phenylethylamine: &1 and &2, each centre independently R or S — all four stereoisomers, i.e. two

diastereomers, each as a racemate. Both centres in &1 would claim only the drawn diastereomer and its mirror image.



Enantiopure + racemic. L-Alanine (abs, S) acylated with racemic 2-phenylpropanoic acid (&1): (S,S) and (R,S), two diastereomers 1:1 — not mirror images of each other, so they usually differ in NMR and can often be separated by ordinary chromatography.

Reagent table under the editor: the same table as a grid below the sheet (scroll the page down to see it) — an alternative to the table on the sheet. Its **Copy for Excel** button copies the table, ready to paste into Excel.

Text



Bold, italic, underline, alignment of multi-line text, subscript and superscript.

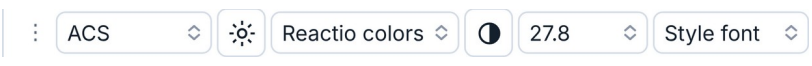
Formula style: digits in the text are written as subscripts (H2SO4 → H_2SO_4).

Count text as a compound: the selected text (NaHCO3, THF…) gets a column in the reagent table.

Count text as a structure: the text stays as written (MeMgBr, K2CO3, nBuLi), and the structure it stands for goes with it: its formula and mass in the reagent table, and the reagent in the reaction SMILES. Known names and formulas are counted without the button; the button is for the others, and a click again removes the structure. **Shift+click** draws the structure in place of the text; a salt (K2CO3) stays one compound, its ions kept together.

Single characters. Double-click a text or an atom label to edit it. The editor shows the result itself, not codes. Select some characters and press **Ctrl+B**, **Ctrl+I** or **Ctrl+U** (⌘ on a Mac), **Ctrl+=** for subscript or **Ctrl+Shift+=** for superscript, or pick a font — only those characters change, e.g. the t of tBu in italic. Shift+Enter starts a new line, Enter finishes.

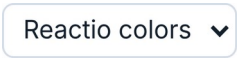
Style





Drawing style of the selected structures, or of the document: ACS, RSC, Nature, Angewandte, Synthesis / Synlett and more.

Edit the style: font, sizes and widths in pt or mm — make your own style. Settings → New documents makes it the style of every new document.



Colour palette of the document (Reactio colors, Publication, Presentation, Pastel, Jmol, Bright, Neon, Jewel, Color-blind safe and your own). **Color-blind safe** uses the eight colours of Okabe and Ito (vermillion, orange, yellow, bluish green, sky blue, blue, reddish purple, black), which people with any kind of colour blindness can tell apart; journals such as Nature recommend this palette for figures.


Paper colour: white, grey, dark grey, night blue, black or blueprint. In Settings the interface can follow the paper, so on a dark paper all panels turn dark too.



Size and font of the selected text or labels.

Colours




Highlight atoms and bonds.




Colour of lines, labels, arrows and text.

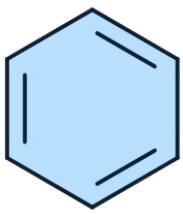




Line width of bonds and arrows (wedges and bold bonds follow).




Ring fill colour.



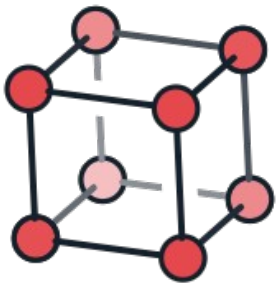
Fill and outline of shapes (circles, rectangles…) and of mark balls: colour, transparency, solid / dashed / dotted / no outline.



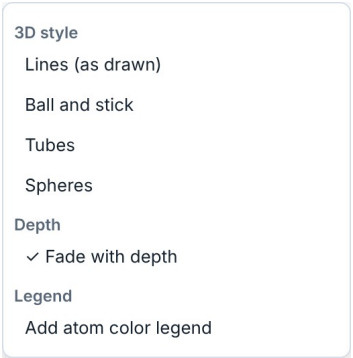
NOSCI **Heteroatom colours** on or off (the whole document; atoms you coloured by hand keep their colour).

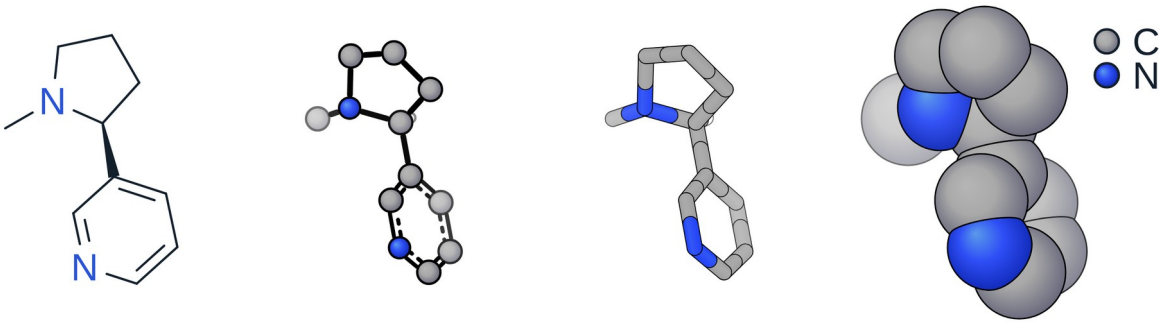
Fade by hand (for the selection) and **automatic fade** (the far side of a 3D structure fades as you turn it). Fade works on 3D structures only.

3D shape: builds a realistic 3D shape (OpenChemLib conformer, then the MMFF94s+ force field). You can also simply start turning a structure with the 3D rotate tool.



3D style: Lines (as drawn), Ball and stick, Tubes or Spheres for the selected structures (with nothing selected, for all of them). The same menu opens on a right-click on a structure. A flat structure gets its 3D shape first. The look follows xyzrender by Alister S. Goodfellow: shaded balls with a black outline, double bonds as two lines, an aromatic ring as one solid and one dashed line, η5 and η6 rings with a dot at the centre and a dashed bond to the metal, spheres that cut into each other. Hydrogens on N, O and S and at stereocentres are drawn; the far side fades with depth.





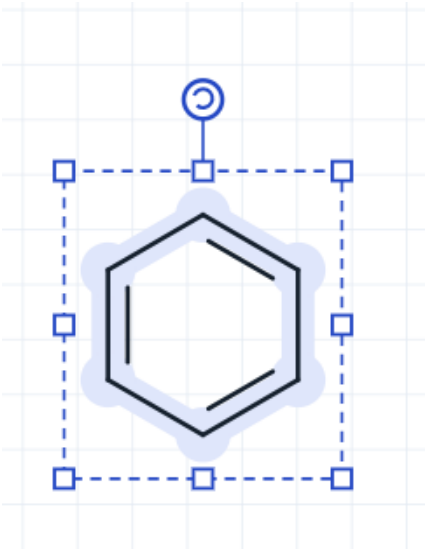
In the same menu: **Fade with depth** on or off for that structure, and **Add atom color legend**: a ball and the symbol of every element, beside the structure, as one group. The style stays when the structure is moved, turned, flipped or copied; editing the structure itself (an atom dragged, a bond added) returns it to lines.

3. Left toolbar

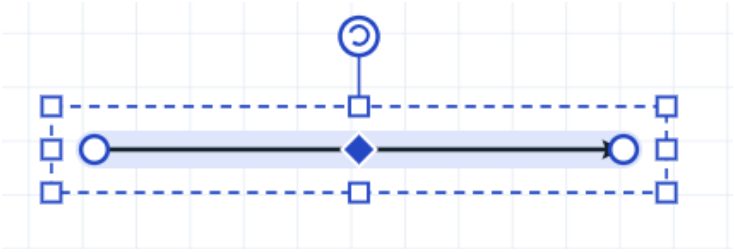
Selection (Space) and **Lasso**. Switching to either selects the molecule you edited last (or, before any edit, the one under the pointer), with its group.

A click selects an atom or bond, a **double-click the whole molecule**, **Shift+click** adds to the selection (or takes out), a box or the lasso selects an area, **Ctrl+A** everything. Dragging an atom that is not selected moves just that atom.

A selection can be restyled with the main toolbar, dragged to move it, and turned, stretched or scaled with the handles around it. Ctrl+drag leaves the original in place and moves a copy; with Shift as well, the copy stays on the same horizontal or vertical line.

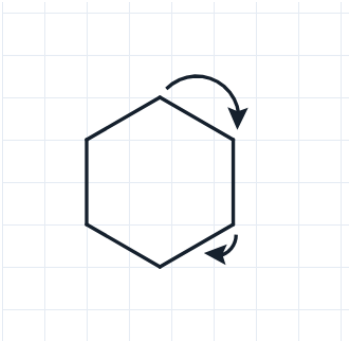


Arrows (E). Click the button again to choose the kind of arrow.

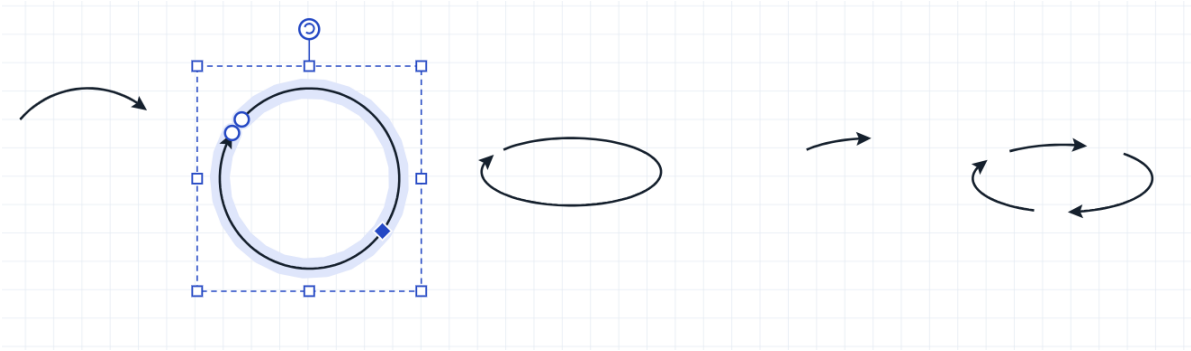


A selected arrow has handles to move, turn and scale it, plus three of its own: the circles at the ends move the start and the end, and the diamond in the middle bends the arrow. Drag the end of a straight arrow to make it longer; with **Shift** it turns in 15° steps; with **Alt** a straight arrow at any angle keeps its direction while you change its length. The end of a curved arrow slides along its own circle or oval (up to a full ring); with **Alt** it moves freely.

While you drag, the arrow itself is shown, faint: the same curve and heads as the one you get. A **curved (mechanism) arrow** starts and ends on an atom or on the middle of a bond: a ring marks the atom or bond it will take. Its ends stay a small gap off the atom or bond (Settings → Drawing → Ends of curved arrows: touching, or a larger gap); at an atom with a label it stops short of the letters.



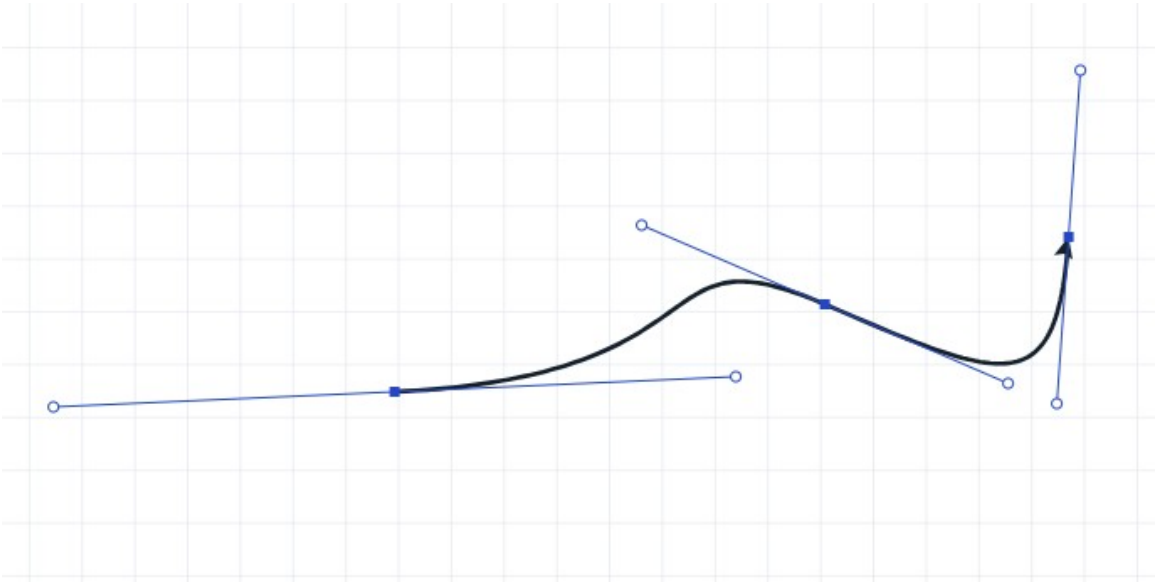
A curved arrow bent into a full circle can be selected and squeezed into an oval; its length then follows the oval. **Ctrl+drag on an end** makes a copy of the arrow in the same place — pull its end (without Ctrl) to build, for example, a catalytic cycle.



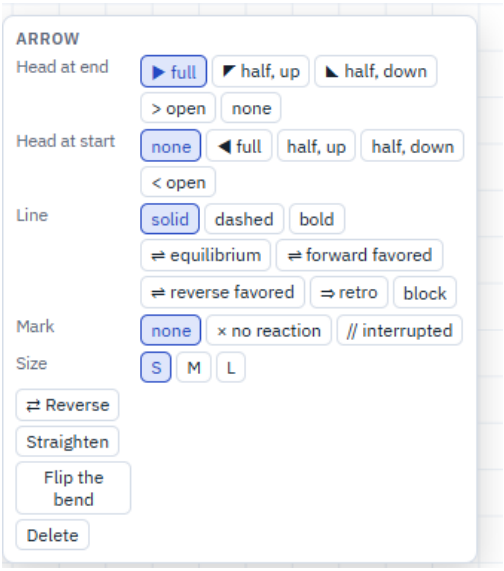
Straight arrows work the same way, for a chain of reactions.



Pen: arrows of any shape. Each click places a point; drag while placing to bend the curve. The points and their handles can be moved later: click the arrow with the pen again.



Right-click any arrow to change its kind (heads, equilibrium, retrosynthetic, dashed, no reaction…).
Double-click an arrow to select the whole reaction with its reactants, products and conditions.



A✚B **Reaction plus**: a + between compounds of a scheme. It separates compounds and is not a charge.

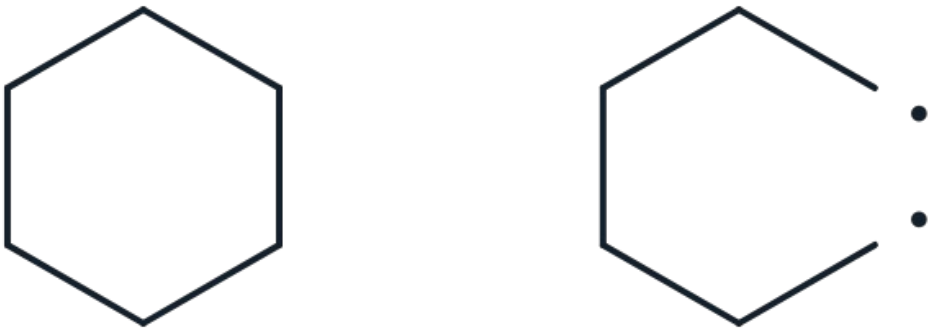
T **Text** (T): click the sheet to write a caption. Text that is a formula (NaHCO3, Pd(PPh3)4) can become a compound of the reagent table; start or end a bond on it with the bond tool and it becomes a label that takes bonds like any substituent, still counted by its formula. Clicked next to the end of a chain, a group such as OEt goes onto that atom as its label.

Eraser: a click removes an object (an atom left with no bond goes with it, so no lone CH4 is left behind); dragging with the left button removes everything in the rectangle; dragging with the right button removes everything the pointer touches.

Hand: moves the view. The middle mouse button does the same with any tool. The mouse wheel scrolls the sheet up and down, Shift+wheel scrolls it sideways (on a touchpad: two fingers in any direction); Ctrl+wheel zooms.

3D rotate: turns a structure in space. A flat structure is first given a 3D shape, which may take a few seconds for a large one; 3D structures turn at once.

Break bonds into radicals: draw a line across the bonds.



Retrosynthetic disconnection: draw a line across the bonds; the pieces become the precursors of a retrosynthetic arrow.

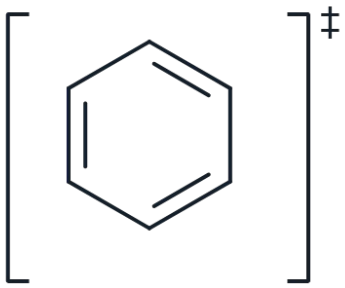


Reaction from a disconnection: the same, as a forward reaction.

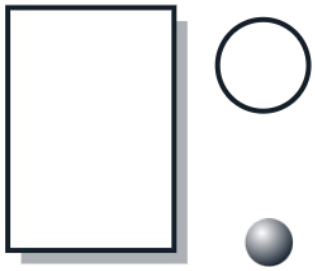


Brackets with their symbols (‡, n, charges…). The last one, ⌉, is the charge corner for the charge of a whole species: drag from where the top line starts: the corner comes out square, with equal sides; stretch it with Select if needed, and put the charge outside the corner.

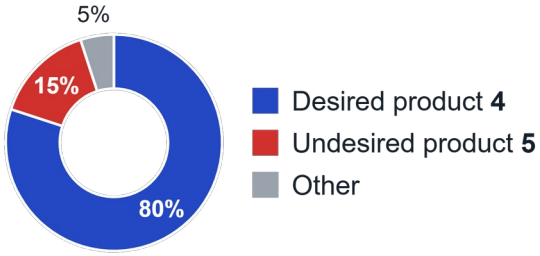


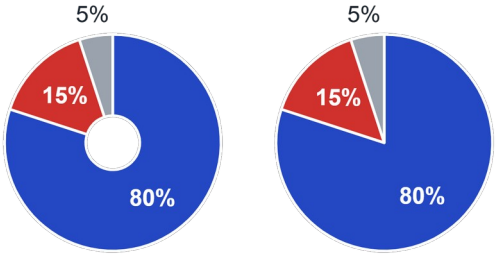


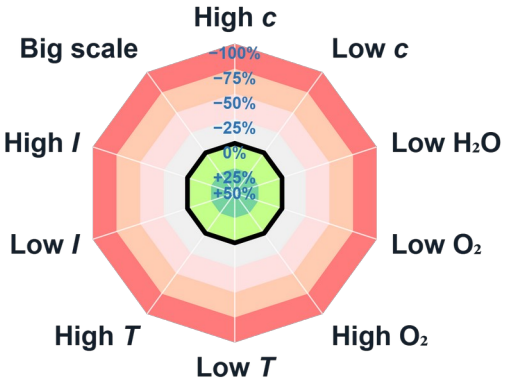
Shapes and lines: circles, ellipses, rectangles, lines and arcs. Click the button again to choose the shape.

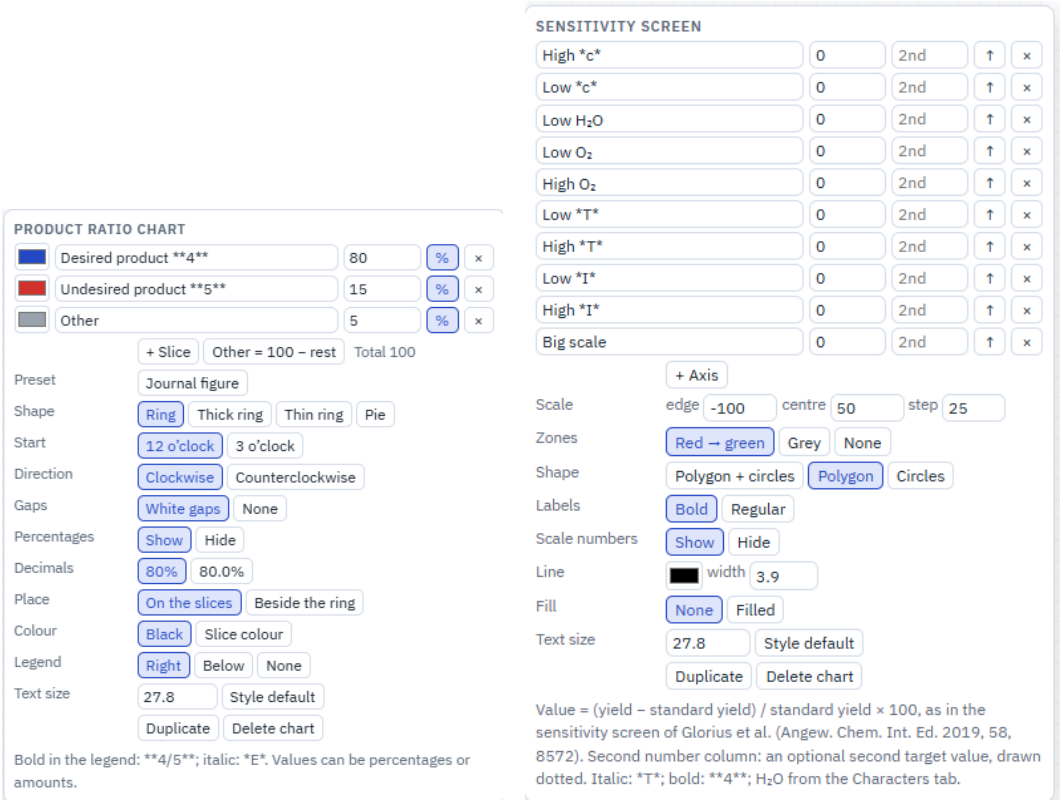


Charts: product-ratio pie or ring charts and sensitivity screens (the radar diagrams of Glorius and co-workers). Click the button again to choose; double-click a chart to edit it.



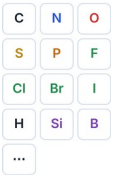








Bonds: a click on the sheet places a bond, dragging pulls it out in 15° steps, a click on a bond with a bond tool makes it that type; a click on a bond that already is that type steps it on (single tool: single ↔ double; double tool: double → triple; triple tool: triple → single). Three buttons with the same set of bond types (single, double, triple, wedges, hashed, bold, dashed, wavy, delocalized). There are three so that you can switch, for example, between wedge and hash with the mouse alone. Over a bond, keys change it: 1, 2, 3 set the order, and **F** puts the bond in front where bonds cross — the bond behind is cut there.

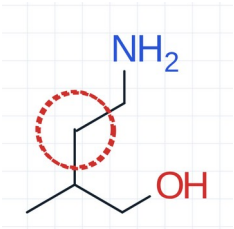


Atoms and the periodic table (…). Over an atom, a key sets the element (C, N, O, S, P, F; L for Cl, B for Br…). A **double-click on a carbon** repeats the last atom or label, however it was put on (typed, by a key or with an element button).

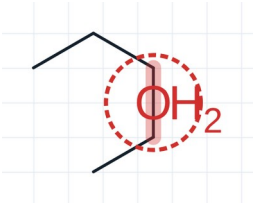
Valence check. After every change the valences are checked: an atom over its normal valence, or left short of it (O instead of OH), gets a **red dashed ring** unless a charge or a radical explains it. Carbenes, nitrenes, nitrenium ions and similar species pass. The rings show only on screen, never in copies or exports; Settings → Warnings → **Mark valence errors on the structure** turns them off. Hypervalent atoms follow RDKit and Marvin: P up to 5 bonds, S and Se up to 6, Cl, Br and I up to 7; a charged atom is checked like the element with the same number of electrons, so PF_6^-, AsF_6^-, BF_4^- and SiF_6^{2-} pass, and neutral P with four bonds takes an H ($HP(O)(OH)_2$, phosphorous acid). A bond can always be added, even past the valence: the atom is then marked.

Structure checks. Run together with the valence check and marked on the structure: **red** for errors (atoms on top of each other, an atom lying on a bond it does not belong to, a stray atom left inside another structure), **orange** for stereo warnings (a wedge or hash at an atom that is not a stereocentre, wedges and hashes that contradict each other), **grey dashes** for hints (an unusually short or long bond, a double bond drawn straight so E/Z cannot be read, two wedges of one kind at a centre with three bonds); orange also marks a wedge between two stereocentres, a wedge on a ring bond where a bond outside the ring could carry it, and a wedge on a double bond (not read as stereo). The stereo rules follow the IUPAC recommendations for drawing stereochemistry (2006, ST-1). A line under Properties counts them; click a count to select the atoms. Settings → Warnings → **Mark stereocentres drawn**

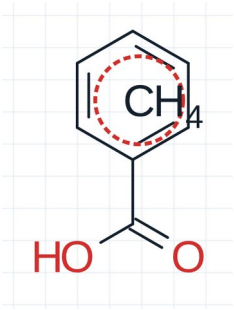
without a wedge or hash adds one more hint (off by default). Like the valence rings, these marks never leave the program.



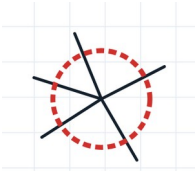
Atoms on top of each other (red): a second molecule dropped onto the first.



Atom on a bond it does not belong to (red): an O lying on a C–C bond.



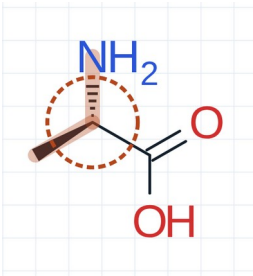
Stray atom inside another structure (red): a lone carbon left in a ring.



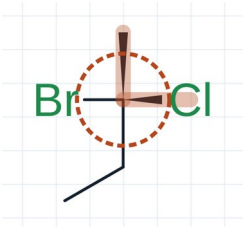
Valence error (red ring): a carbon with five bonds.



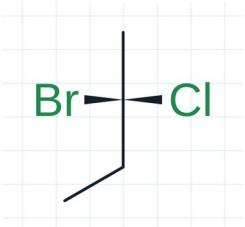
Wedge not at a stereocentre (orange): the CH carrying two identical methyl groups is not a stereocentre.



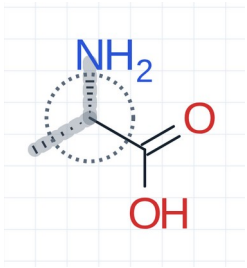
Contradicting wedges (orange): a wedge and a hash at alanine's centre that give opposite configurations (IUPAC ST-1.2.10: not acceptable). Draw one wedge or one hash.



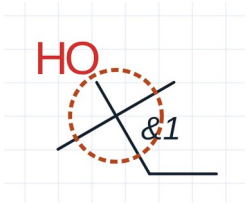
Two wedges on neighbouring bonds (orange): at a centre drawn with all four bonds, two wedges side by side give different configurations (IUPAC ST-1.1.11: ambiguous, never use).



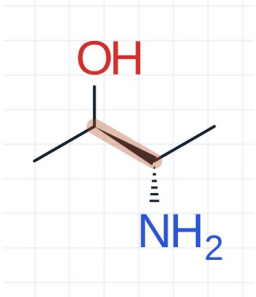
The same centre, acceptable: wedges on the two opposite bonds (Br and Cl toward the viewer, the carbons away). No mark. IUPAC accepts this (ST-1.1.3) but prefers one wedge, or one wedge and one hash on neighbouring bonds.



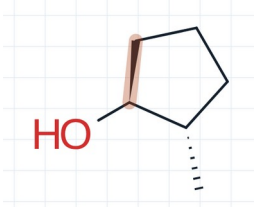
Two hashes at a centre with three bonds (grey hint): clear — both say the H points to the viewer — but IUPAC prefers one (ST-1.2.5).



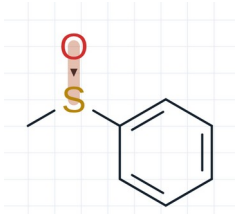
Stereo group on an atom that is not a stereocentre (orange): &1 on a carbon with two methyl groups.



Wedge between two stereocentres (orange): the wedge joins two stereocentres, so it is unclear which one it describes (IUPAC ST-0.5: avoid).



Wedge on a ring bond (orange): put it on the OH instead (IUPAC ST-0.2: not acceptable when a bond outside the ring could carry it).



Wedge on a double bond (orange): a chiral sulfoxide is fine, but its wedge belongs on a single bond (S–CH_3 or S–Ph; IUPAC ST-1.5). A wedge on S=O is not read as stereo: SMILES, InChI and MOL keep the double bond without configuration.



Unusually long bond (grey hint): one bond much longer than the others.



Double bond drawn straight (grey hint): E/Z of but-2-ene cannot be read.



Symbols: two buttons with charges, radicals, lone pairs, attachment points, the mark ball and partial charges δ+ / δ− — two, so that + and − are both at hand.

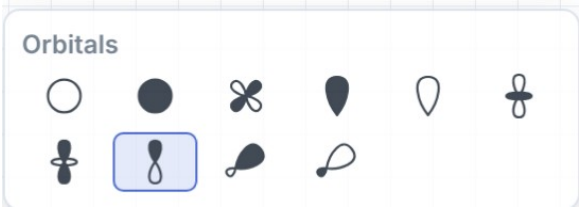


Partial charges δ+ and δ−: click an atom, again to remove; drag the symbol to move it. Drawing only: formula and valence do not change.

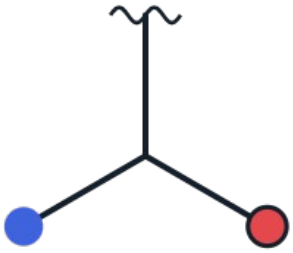


Orbitals: s (empty or filled), dxy, a single lobe (filled or empty), dz^2 (the ring or the lobes filled), p, and sp^3 (the big or the small lobe filled). Click an atom (or drag to set the direction, in 15° steps); click again for the orbital menu. All orbitals are drawn at one size (Alt+drag sets the size too). An orbital put on an atom stays on it when the atom moves. Selected, it has two handles: the **round** one turns it about its atom (the size stays; Shift for 15° steps), the **square** one makes it bigger or smaller (the

direction stays). With the bond tool, drag from one orbital to another for the overlap line: a line starts and ends at the tip or the middle of a lobe (all four of a d orbital, both of an sp^3), or at the centre of an s orbital or anywhere on its circle; the dashed bond tool gives a dashed line. Orbitals go to and come from CDX and CDXML files.



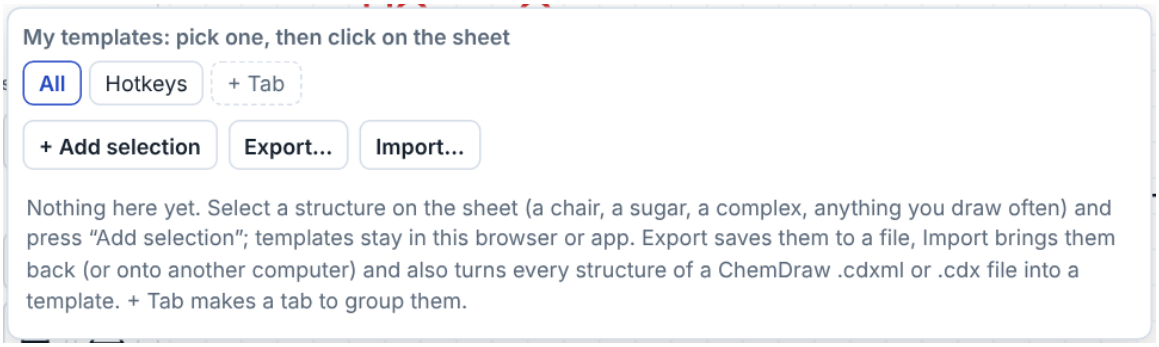
 **Mark ball**: a light ball on an atom for highlighting (each click makes it bigger, then removes it). Its fill and outline are set with the fill button.



 **Rings** of any size, also your own (for example 18). Click a bond to fuse the ring onto it, an atom to attach it (inside a chain or ring: a spiro ring), or empty sheet for a ring on its own.

 **Bridged** ring systems (norbornane, adamantane, cubane…).

 **My templates**: select a structure and press **Add selection** in this menu. Templates can be grouped in tabs (make a tab and drag templates onto it). In the Hotkeys tab of the menu, every template can get its own key.



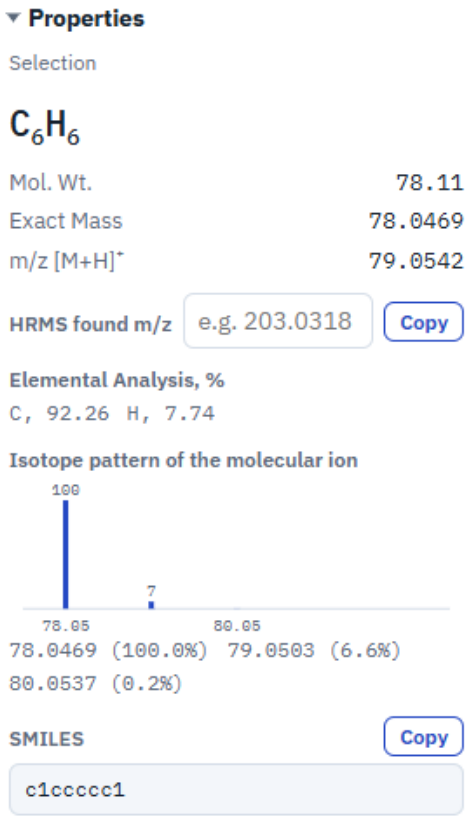
Import reads two kinds of files. A Reactio Draw templates file (**.json**, made with Export) brings its templates back as they were saved, in their own tabs. A **.cdxml / .cdx** file turns every structure in it into a template, scaled to the sheet’s bond length, on a new tab named after the file; a salt (cation and anion drawn together) stays one template.

4. Right panel

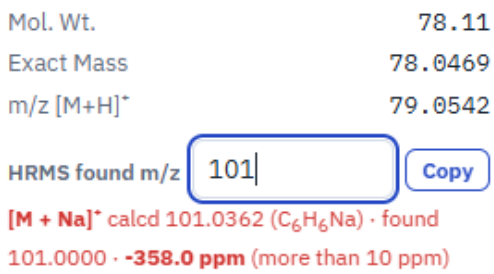
Five tabs: **Analysis, Characters, Labels, Settings, Info**. The panel can be made wider or narrower by dragging its left edge. The arrow at the left of the tabs folds the panel to a thin strip at the right edge, so the sheet gets the room; the arrow on the strip opens it again.

Analysis

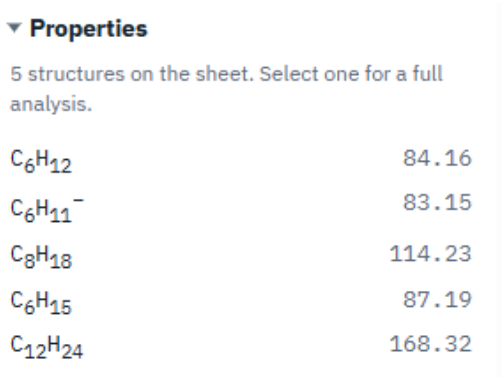
With one structure on the sheet (or one selected) it shows the formula, molecular weight, exact mass, m/z $[M+H]^+$, elemental analysis, the isotope pattern, SMILES, InChI and InChIKey.



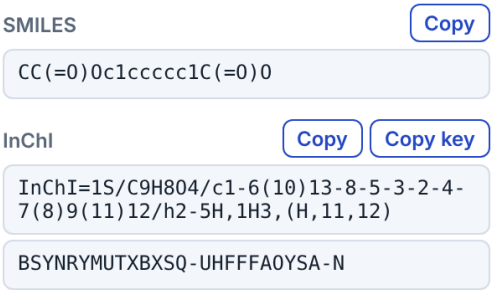
HRMS. Type the m/z found in the mass spectrum: the program suggests the ion (for example $[M+H]^+$, $[M+Na]^+$) and gives the error in ppm; above 10 ppm the line turns red.



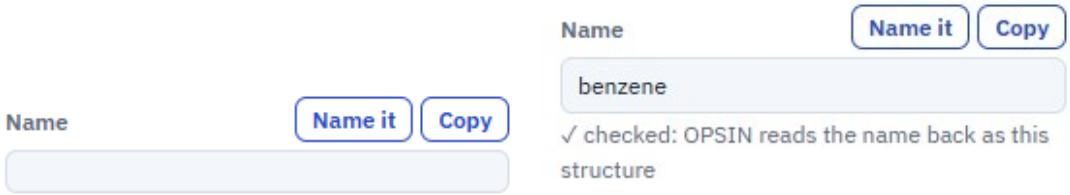
With several structures the tab lists them. Select one, or click its formula in the list, to see its data.



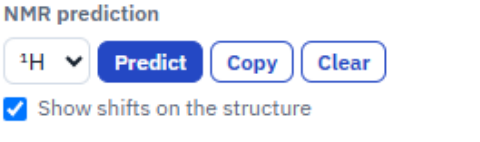
SMILES, InChI and InChIKey. SMILES is canonical: the same structure gives the same SMILES however it was drawn, with the drawn stereochemistry. InChI and InChIKey come from the official InChI library (InChI Trust / IUPAC), built into the program: they work offline and in the browser. Copy and Copy key put them on the clipboard.



Name it (Mac, Windows and Linux apps) makes the IUPAC name with openclatura and checks it by reading it back with OPSIN; a name marked ✓ was read back as the same structure. openclatura names organic structures only: inorganic species (PF_6^-, BF_4^-, ClO_4^-, SO_4^{2-}, metal ions and complexes, water) get no name. A salt is named part by part — the organic ion gets its name, the inorganic one shows as “?” (for example “1-methylpyridin-1-ium; ?”) — so a full salt name such as “1-methylpyridinium hexafluorophosphate” has to be written by hand. Trust only names marked ✓.



NMR prediction: ^1H, ^{13}C, ^{19}F and ^{31}P shifts from measured spectra (nmrshiftdb2). The shifts appear as a stick spectrum and a table, and can be shown next to the atoms; pointing at an atom lights its row in the table (and a row lights its atoms); **Clear** takes them off. In the web version the tables (2.2 MB) are downloaded on the first Predict, after asking. **Copy** gives a line for the supporting information, ending with a sentence naming the data source (nmrshiftdb2 and its licence). It is a prediction, not a measured spectrum.



NMR prediction

¹H Predict Copy Clear

Show shifts on the structure



7.09	±0.39	1H	sphere 2, 76 ref.
6.99	±0.23	1H	sphere 3, 13 ref.
6.90	±0.39	1H	sphere 2, 78 ref.
2.30	±0.23	3H	sphere 3, 31 ref.
2.22	±0.17	3H	sphere 4, 8 ref.
2.19	±0.17	3H	sphere 4, 12 ref.

A counter-ion written as text (PF6−, BF4−) counts in the formula and masses when it is selected with the structure, or always when it stands beside the only structure on the sheet.

Labels on the sheet. Select a structure, tick the data you want (formula, MW, exact mass, m/z, elemental analysis) and press **Paste**: the lines go under the structure and update themselves.

Labels on the sheet

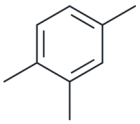
Paste under the structure

Formula — Exact Mass

Mol. Wt. — m/z

Elem. Anal. — Decimals 4

Paste



Chemical Formula: C_9H_{12}
Exact Mass: 120.0939
m/z: 120.0939 (100.0%), 121.0973 (9.9%)

Compound numbers. Select the structures and give a template: {n} gives 1, 2, 3…, {a} gives a, b, c… (4.{n} → 4.1, 4.2; S{n} → S1, S2; 3{a} → 3a, 3b), and the first number. The number is bold; the second box (**then, not bold**) adds text after it in plain type — 5.{n} with (xx%) gives **5.1** (xx%), and you only have to replace xx with the yield. Numbering again replaces the numbers. The numbers then **keep themselves in order**: number a new compound between two others and the later numbers move on; delete a structure and its number goes. Each series (a template with its first number: {n} from 1, 3.{n} from 1) is counted on its own, and a copied structure takes its number along. Type a letter after a number (2a, 2b) and those compounds share one number.

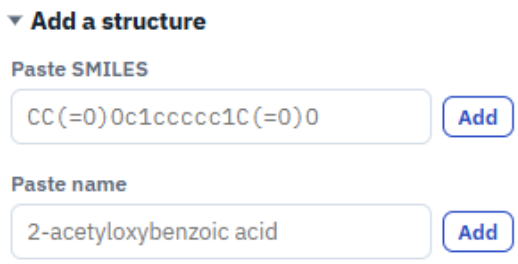
Compound numbers

5.{n} from 1 Number

then, not bold (xx%)

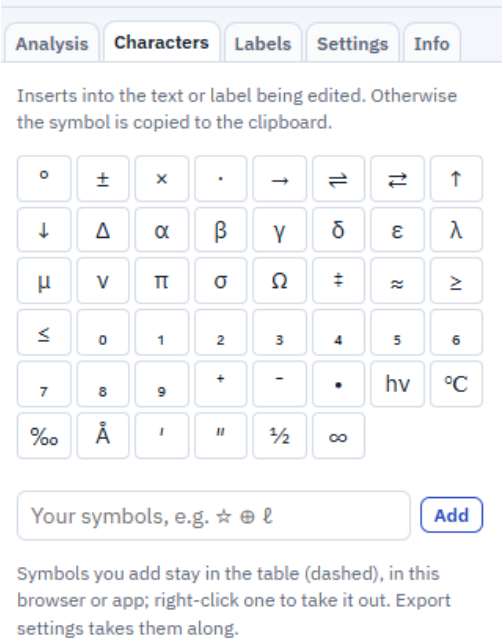
{n}: 1, 2, 3... {a}: a, b, c... Examples: 4.{n} → 4.1, 4.2; S{n} → S1, S2; 3{a} → 3a, 3b. The second box is written after the number in plain type: (85%) gives **4.1** (85%). Numbers the selected structures; numbering again replaces their numbers.

Add a structure from a SMILES or a name. About 600 names and abbreviations (solvents, reagents, catalysts, amino acids, sugars, drugs; the list includes the reagents of the Beilstein Chemical Lookup Collection) work everywhere; other systematic names are read by OPSIN in the apps.



Characters

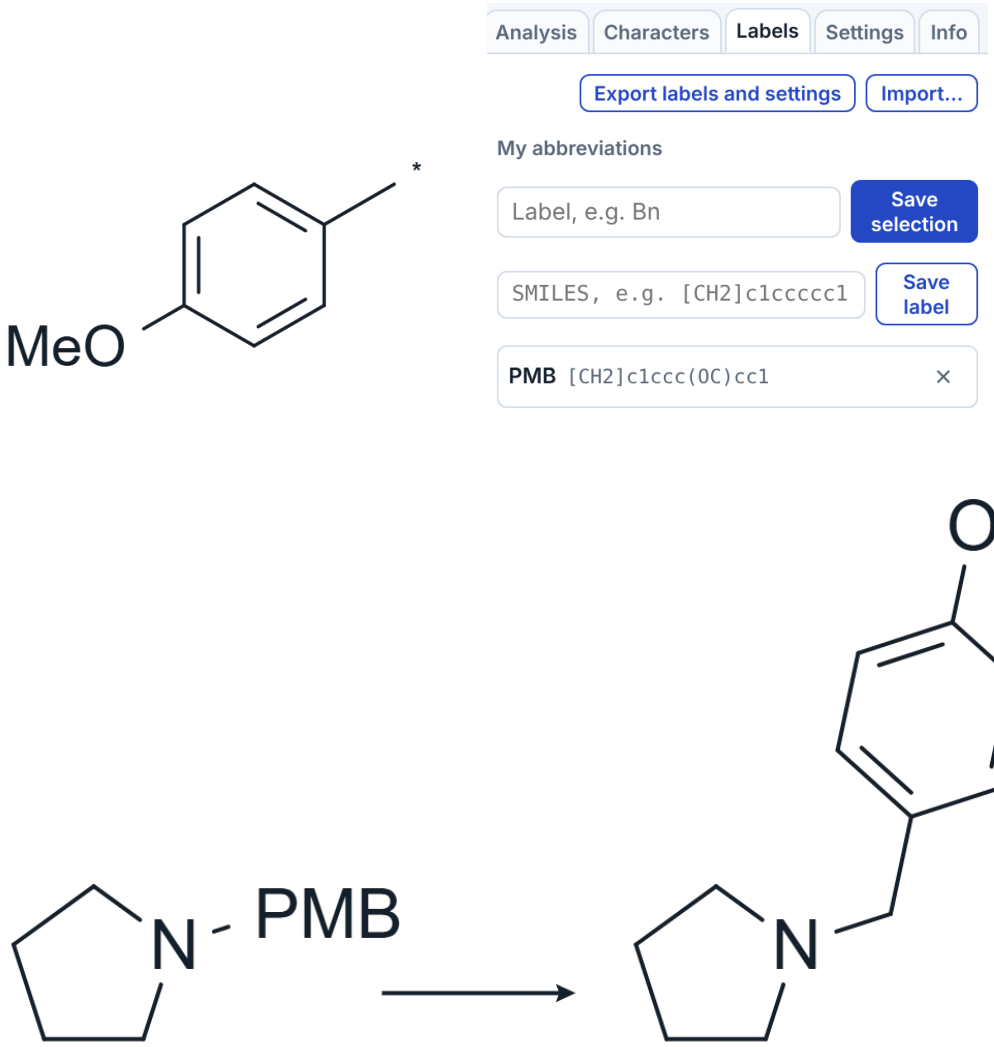
Symbols for text: °, ±, →, ⇌, Δ, Greek letters, sub- and superscript digits, hν, Å and more. A click inserts the symbol into the text being edited, otherwise copies it. **Add your own** in the box under the table; they stay, with a dashed frame; right-click one to remove it.



Labels

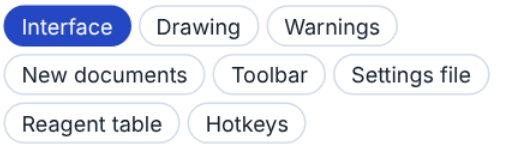
Labels are read by an algorithm, not looked up in a list: CO2Me, SO2Me, CONH2, Si(iPr)3, SnBu3, (CH2)3OH, P(O)(OEt)2, N(Me)Boc and the like are understood as they are typed, and formula, mass and SMILES count them. Group names (Me, tBu, Ph, Boc, Ts, TBS…) are the only list; it includes the abbreviations of the Beilstein Chemical Lookup Collection (OTES, DMTr, 2-py and about 200 more). A label written as a formula with no single structure (C10H7) is counted by its formula, and so is a label with a metal, however many bonds its atom has (Co(CO)3 in the cobalt complex of an alkyne). A label that cannot be read stays red. When the bond comes from the right, a label is written backwards: OEt → EtO, Si(iPr)3 → (iPr)3Si.

Your own abbreviations. Draw the group and put an attachment point  (Symbols) on the atom where it attaches. Select the group, type the label (for example PMB) and press **Save selection**. From now on PMB is understood in structures, and Expand Label  draws it out. A label has one attachment point.



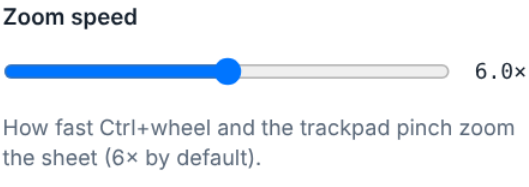
Settings

The settings are split into small tabs at the top of the Settings tab, one group shown at a time (the last one opened comes back): Interface, Drawing, Warnings, New documents, Toolbar, Settings file, Reagent table and Hotkeys.

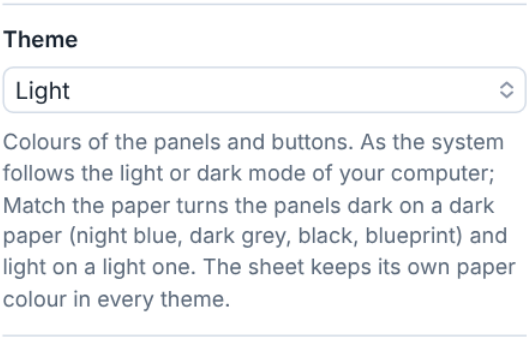


Interface

Zoom speed: how fast Ctrl+wheel and a pinch on the trackpad zoom the sheet (6× by default).



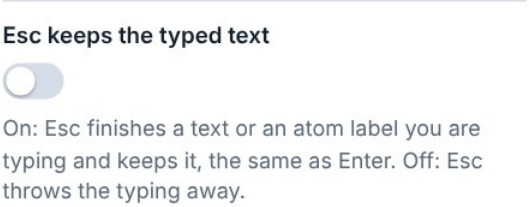
Theme of the panels and buttons (light at first): as the system (light or dark like your computer), light, dark, or matching the paper (dark panels on a dark paper, light on a light one). The sheet keeps its own paper colour in every theme.



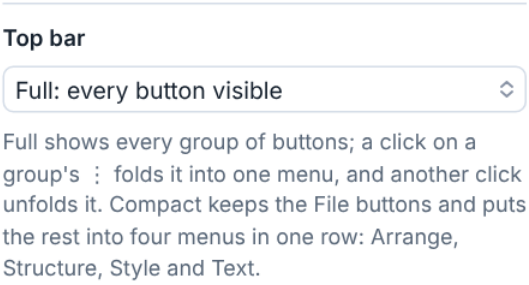
Accent colour of buttons, selection and highlights: Blue, Teal, Green, Violet, Crimson, Amber, Graphite, or your own colour. The drawing keeps its own colours; a colour too light or too dark to read is adjusted for each theme.



Esc keeps the typed text: on, Esc finishes a text or an atom label like Enter; off, Esc throws the typing away.



Top bar: Full shows every group of buttons (a click on a group's grip ⋮ folds that group into one menu); Compact keeps the file buttons and puts the rest into four menus in one row.



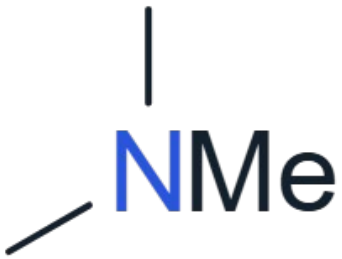
Drawing

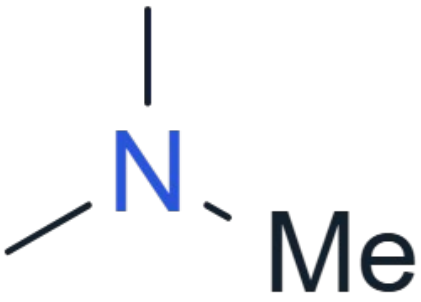
NMe, NBoc, SiMe2 typed on an atom in a chain or ring: one label (the group beside the atom) or the group on a bond of its own. Typing it with a dash (N-Me) always gives the bond.

NMe, NBoc, SiMe2 typed on an atom in a chain or ring

One label: the group beside the atom (N ov

How an element with a group, typed on an atom that already has bonds, is drawn. Typing it with a dash (N-Me) always puts the group on a bond.



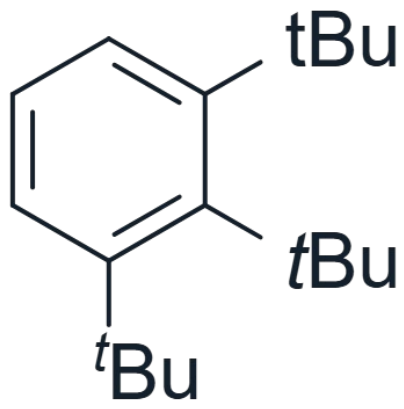


t, i, s, n in groups (tBu, iPr, sBu, nBu): plain, italic, or italic superscript. New labels follow the choice; **Apply to selection** changes the selected structures (or the whole sheet).

t, i, s, n in groups (tBu, iPr, sBu, nBu)

Leave plain: tBu | Apply to selection

How the prefix letter of tBu, iPr, sBu, nBu, OtBu, CO2tBu is written. New labels you type follow it; Apply to selection changes the labels of the selected structures (the whole sheet when nothing is selected).



Ends of curved arrows: the ends of a curved arrow stay a small gap off the atom or bond (at first), a larger gap, or touch it. An atom with a label always keeps the arrow clear of its letters.

Ends of curved arrows

Small gap (default)

How far a curved (mechanism) arrow drawn from or to an atom or the middle of a bond stays off it. An atom with a label always keeps at least a small gap, so the head does not run into its letters. Arrows already drawn keep their ends.

Clean Up: a methoxy drawn as O–CH3: leave it as drawn (a line, the default), or write it as the label OMe or OCH3 (H3CO on the left). A crowded methoxy that Clean Up has to write as a label follows this choice (OCH3 when it is Leave as drawn).

Clean Up: a methoxy drawn as O–CH3

Leave as drawn (a line)

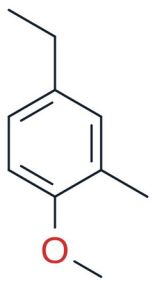
What Clean Up and Clean Up 2D do with a methoxy group drawn out as O and a line. A crowded group that Clean Up has to write as a label follows this too (OCH3 when Leave as drawn).

Clean Up: a methyl drawn as a line end: leave it as drawn (the default), or write it as Me or CH3 — the methyl of toluene, the CH3 of an ethyl; ethane is left alone. Both choices go into the settings file.

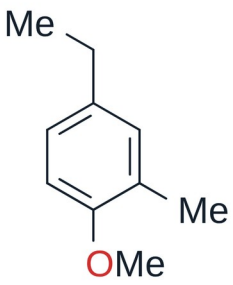
Clean Up: a methyl drawn as a line end

Leave as drawn (a line)

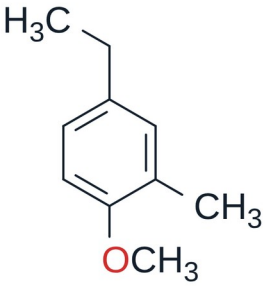
What Clean Up and Clean Up 2D do with a methyl drawn as the bare end of a bond (toluene, the CH3 of an ethyl). Ethane and the methyl of a methoxy written as OMe or OCH3 are left alone.



Leave as drawn



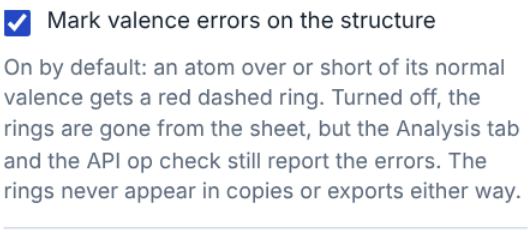
OMe, Me



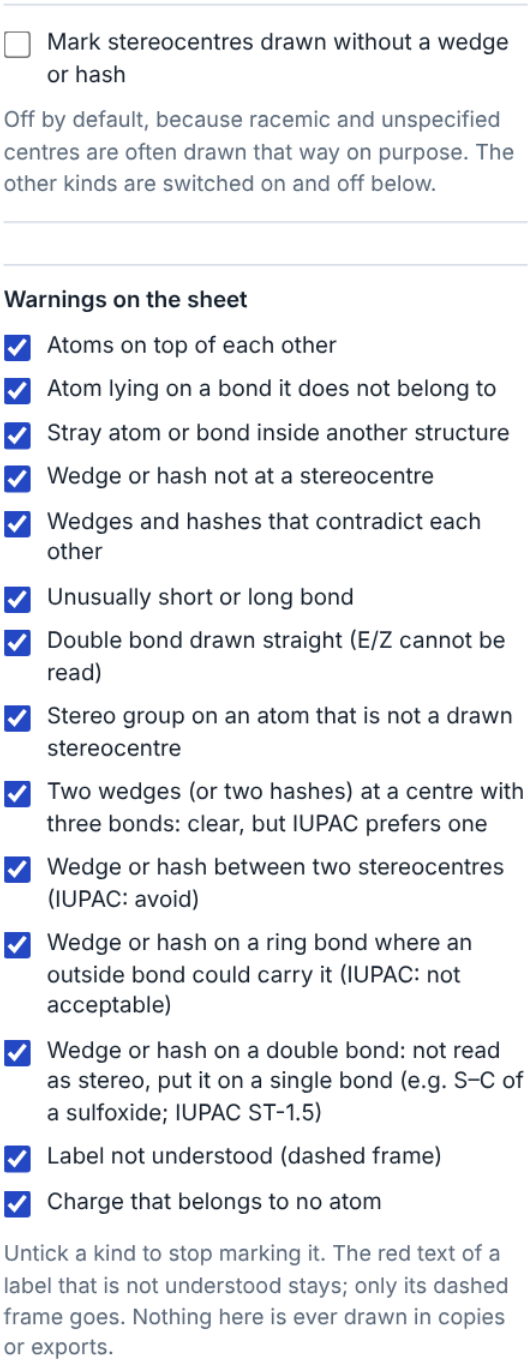
OCH3, CH3

Warnings

Mark valence errors on the structure (on by default): an atom over or short of its normal valence gets a red dashed ring. Turned off, the rings leave the sheet, but the Analysis tab and the API still report the errors. The rings never appear in copies or exports.

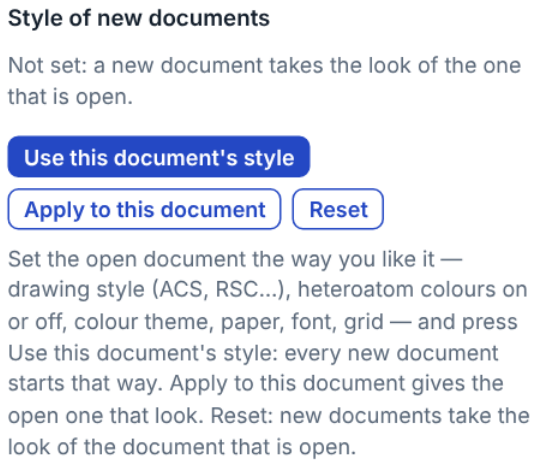


Mark stereocentres drawn without a wedge or hash (off by default, since racemic and unspecified centres are often drawn that way on purpose). The other structure checks show only on screen and are listed in the Analysis tab (see Structure checks in chapter 3). **Warnings on the sheet** has a box for each kind of warning, also for the dashed frame of a label that is not understood and for a charge that belongs to no atom: untick a kind to stop marking it.



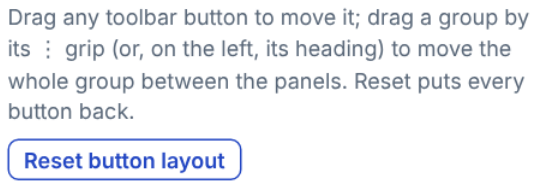
New documents

Set the open document the way you like it (drawing style, heteroatom colours, palette, paper, font, grid) and press **Use this document's style**: every new document starts that way. **Apply to this document** gives the open one that look; **Reset** lets new documents take the look of the one that is open.



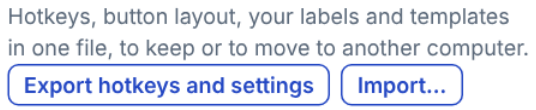
Toolbar layout

Drag a button to move it, or a group by its grip ⋮ (on the left, by its heading). **Reset button layout** puts every button back where it was.



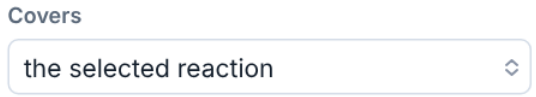
Settings file

Export hotkeys and settings saves hotkeys, button layout, accent colour, your labels, palettes, symbols and templates in one file, to keep or to move to another computer or colleague; **Import…** reads it back.



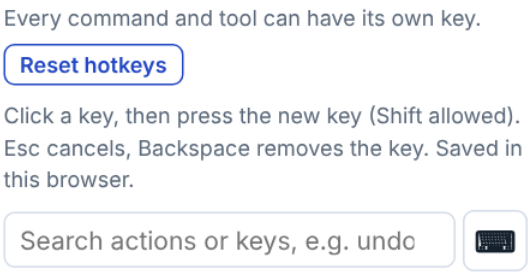
Reagent table

Covers: the whole sheet (every compound on the sheet, for a document with one reaction) or the selected reaction (select the scheme, then make the table; structures drawn later are not added). See Reagent table in chapter 2.

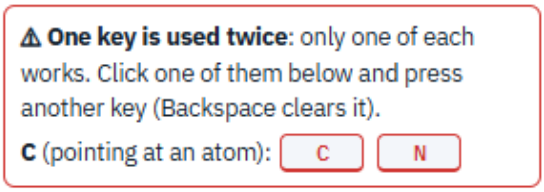


Hotkeys

Every command, tool and template can have its own key: we do not know which keys you like, so everything can be set. Search by name, or press ⌨ and then the keys to find what they do. Click a key, then press the new one; Esc cancels, Backspace removes the key. **Reset hotkeys** brings back the original keys.



A key used twice is caught: a red dot on the Settings tab, a box at the top of the hotkeys list with each pair in red, and a warning. Only the same key in the same place counts — both among the commands, or both while pointing at an atom; a command key and an atom key may be the same.



At the end of the tab are two lists that cannot be changed: the **fixed keys** (Esc, Delete, the arrow keys that walk the hotspot or nudge the selection, Alt+← → to turn it by 15°, Ctrl+C / Ctrl+V…) and what the **mouse** does (a ring or template on a bond fuses it, Ctrl+drag copies, Shift+drag moves along one axis, the round handle rotates…).

Info

The coordinates and angles of the selection, the switches of the API (chapter 6), the **version** of Reactio Draw with a link to what is new in it, and the credits: the components the program uses and their licences.

5. Drawing fast from the keyboard

- **Space** switches to Selection and selects the molecule you edited last.
- **Ctrl+→** with a structure selected draws a reaction: an arrow and a copy of the structure as the product — only the changes are left to draw.

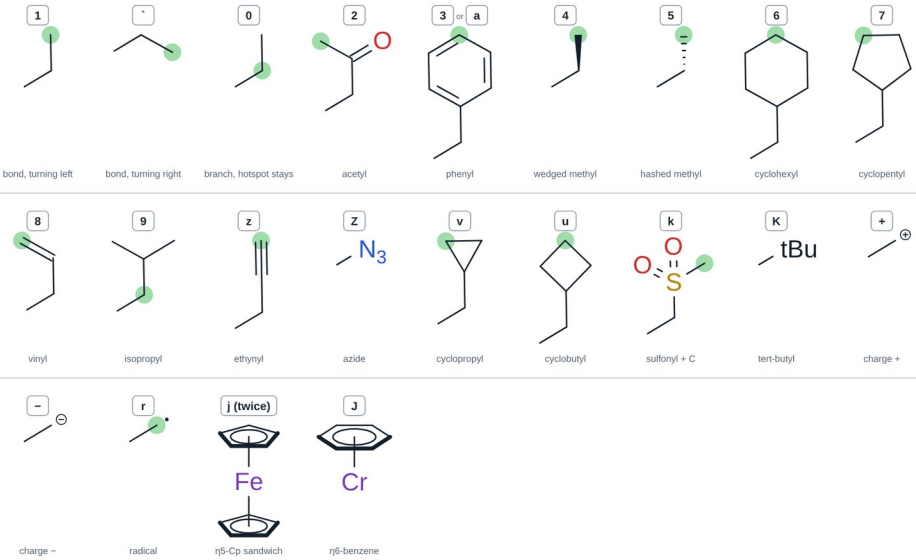


- **The hotspot.** The atom or bond under the pointer lights up and stays the hotspot (green) when the pointer moves away. The arrow keys move it along the structure (atom → bond → atom); Shift+arrow goes to the last atom that way, along the chain or straight across a ring, Alt+Shift+arrow to the next structure. Enter opens the label of the hotspot atom for typing.
- **Keys over an atom** add groups (the hotspot moves on to the new group) or change the atom (CH3 → OH…); on a middle carbon they add C=O, two methyls or a spiro ring. **Keys over a bond** fuse a ring or change the bond (double, wedge…). All of them are in the file Reactio Draw - Hotkeys.reactio, a one-page chart with each key drawn as a keycap beside what it draws (File → Save as turns it into CDXML for other editors).

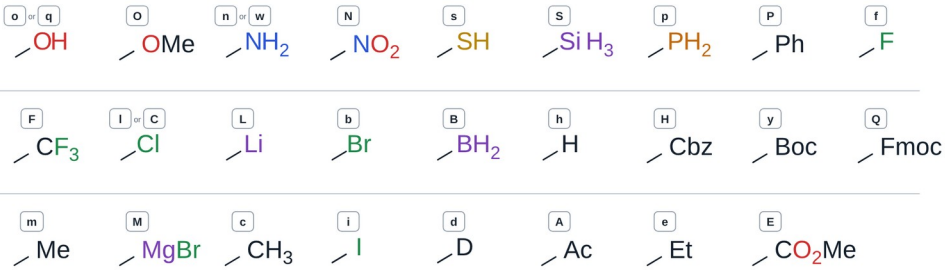
Reactio Draw hotkeys

Point at an atom or a bond and press a key. Green marks the hotspot, where the next key acts. A capital letter means the key with Shift.

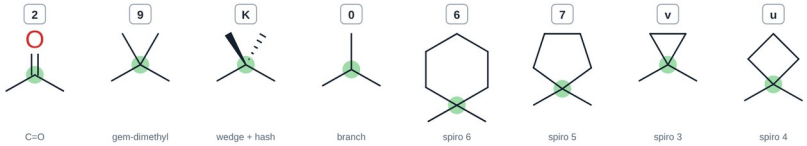
Atom at the end of a chain



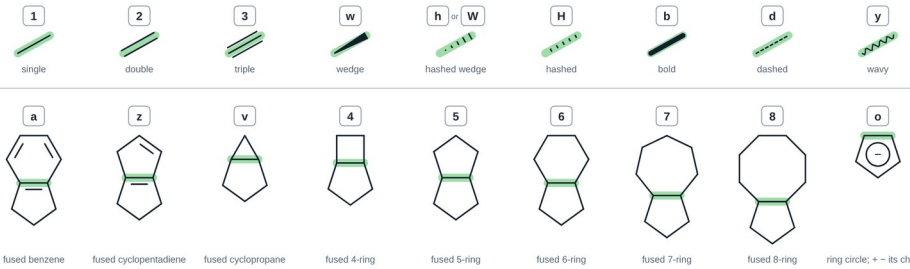
Labels and elements on that atom



Atom in the middle of a chain



Bond



Moving around

← → ↑ ↓	move the hotspot: atom, bond, atom over the structure
Shift+arrow	to the last atom that way: along the chain, or straight across a ring
Alt+Shift+arrow	to the next structure
Enter	edit the label
[then]	bond two atoms (ring closure): [on the first,] on the second
Ctrl+Shift+L	two atoms selected: bond them and clean up
Esc	no hotspot (a click on empty sheet does the same)
, or .	one hydrogen fewer, one more
l or r or c	on a double bond: second line left, right or centred
f	bond in front at a crossing

Every key can be changed in Settings → Hotkeys.

- With two atoms selected, **Ctrl+Shift+L** bonds them and cleans up the structure.
- **,** and **.** take one hydrogen off or add one.

6. Working with the API

Reactio Draw can be driven by commands: draw a structure from a SMILES or a name, read the structures back with their formula, masses, SMILES, InChI and NMR shifts, fill the reagent table, export the drawing. The same commands work in three ways:

Way	Who uses it	Where
Local API	scripts on your computer (Python, R, curl, an ELN tool)	Mac, Windows and Linux apps; HTTP on 127.0.0.1
Page API	code running in the page itself (the browser console, a userscript)	every version: `ReactioAPI.run(op, args)`
Embedding	a web site that shows the editor in an iframe	web version with ?embed; postMessage
Command line	you or a script in a terminal, for files: pictures, conversion, checks	Mac, Windows and Linux apps: `reactio`; no window opens

Every command has a name (`op`) and arguments (`args`, an object). The answer is a result object, or an error with a message. SMILES in answers are canonical and carry the drawn stereochemistry (@/@@ from wedges and hashes, / and \ for E/Z).

6.1 Local API (desktop apps)

- **Turn it on**: Info tab → **Local API**. It is off until you turn it on; the choice is remembered; turn it off and nothing listens.
- **Where it listens**: http://127.0.0.1:<port>/api — on this computer only, never from the network.
- **Port and token** are in api.json, which exists while the app runs:
  - Windows: `%APPDATA%\ReactioDraw\api.json`
  - Mac: `~/Library/Application Support/ReactioDraw/api.json`
  - Linux: `~/.config/ReactioDraw/api.json`
  - It contains `{"url": "http://127.0.0.1:47615/api", "token": "…", "port": 47615}`; the token changes at every start.
- **Request**: `POST <url>` with the header `X-Reactio-Token: <token>` and a JSON body `{"op": "…", "args": {…}}`. **Answer**: `{"ok": true, "result": …}` or `{"ok": false, "error": "…"}`.
- **Safety**: requests without the right token are refused, and so is anything a web page sends (requests with an Origin header) — a site you visit cannot drive the app. The address must be 127.0.0.1 or localhost with the port from api.json.

Python (no extra packages)

```
import json, os, pathlib, urllib.request

p = pathlib.Path(os.environ["APPDATA"]) / "ReactioDraw" / "api.json"              #
Windows
# p = pathlib.Path.home() / "Library/Application Support/ReactioDraw/api.json"   #
Mac
# p = pathlib.Path.home() / ".config/ReactioDraw/api.json"                         #
Linux
info = json.loads(p.read_text())

def rd(op, **args):
    req = urllib.request.Request(info["url"], json.dumps({"op": op, "args":
args}).encode(),
                                 {"X-Reactio-Token": info["token"], "Content-Type":
"application/json"})
```

```
    answer = json.load(urllib.request.urlopen(req))
    if not answer["ok"]:
        raise RuntimeError(answer["error"])
    return answer["result"]

print(rd("calc", smiles="CC(=O)Oc1ccccc1C(=O)O"))     # aspirin: C9H8O4, 180.159 …
rd("addName", name="(2S)-2-aminopropanoic acid")      # draws L-alanine
print(rd("inchi", index=0))                            # InChI and InChIKey
print(rd("nmr", nucleus="13C", index=0))               # predicted 13C shifts
open("drawing.cdxml", "w").write(rd("export", format="cdxml")["cdxml"])
```

curl

```
curl -s -X POST http://127.0.0.1:47615/api \
     -H "X-Reactio-Token: <token from api.json>" -H "Content-Type:
application/json" \
     -d '{"op": "addSmiles", "args": {"smiles": "c1ccccc1O"}}'
```

R

```
library(httr); library(jsonlite)
info <- fromJSON(file.path(Sys.getenv("APPDATA"), "ReactioDraw", "api.json"))
rd <- function(op, args = list()) {
  r <- POST(info$url, add_headers("X-Reactio-Token" = info$token),
            body = list(op = op, args = args), encode = "json")
  content(r)$result
}
rd("list")
```

6.2 Page API

In every version — the apps, the downloaded page and the web site — the page has `window.ReactioAPI`. Open the browser's developer console (F12) on the Reactio Draw page, or run code in the page, and call:

```
await ReactioAPI.run("addSmiles", {smiles: "CCO"})
await ReactioAPI.run("list")
ReactioAPI.ops()            // the names of all commands
```

Commands that take time (names, NMR, InChI) return a Promise, so use await.

6.3 Embedding in a web page

A site — an electronic lab notebook, a course page, a database search form — can show the editor and talk to it:

```
<iframe id="rd" src="https://paivel.github.io/reactio-draw/web/index.html?
embed&origin=https://my-site.org"
        style="width:100%;height:640px;border:0"></iframe>
```

- `?embed` shows the editor without its title bar and tabs, always starting with a blank sheet; the site never sees the drawings the visitor saved elsewhere.
- `origin=` names the sites allowed to send commands (several, separated by commas). Without it the editor answers only the page that embeds it.
- Send `{reactio: 1, id, op, args}` with postMessage; the answer comes back as `{reactio: 1, id, ok, result}` or `{reactio: 1, id, ok: false, error}`.
- The editor also sends `{reactio: 1, event: "ready", ops: […]}` when it has loaded, `{reactio: 1, event: "change"}` each time the drawing changes and `{reactio: 1, event: "selection"}` each time the selection changes.
- `?embed&readonly` only shows the drawing: toolbars and panels are hidden, the visitor can move, zoom and copy it but not change it (the command `readOnly` switches this on and off).

```
const frame = document.getElementById('rd');
let next = 0; const waiting = {};
addEventListener('message', e => {
  if (e.source !== frame.contentWindow || !e.data || e.data.reactio !== 1) return;
  const d = e.data;
  if (d.id && waiting[d.id]) { waiting[d.id](d); delete waiting[d.id]; }
});
function rd(op, args) {
  return new Promise((resolve, reject) => {
    const id = ++next;
    waiting[id] = d => d.ok ? resolve(d.result) : reject(new Error(d.error));
    frame.contentWindow.postMessage({reactio: 1, id, op, args},
'https://paivel.github.io');
  });
}
// show a structure, let the user edit it, read it back as a molfile
await rd('set', {text: 'CC(=O)Oc1ccccc1C(=O)O'});
const molfile = (await rd('export', {format: 'mol'})).mol;
```

You can also host the editor yourself: copy the web/ folder (the page and its NMR tables file) to your server and point the iframe at that copy. The details are in EMBEDDING.md with the source code.

6.4 Command line

reactio runs Reactio Draw without a window: it reads the files you name, does the job and quits. The app must be installed, but it need not be open, and your open documents are not touched. Where it is: on Windows **reactio.exe** in the program folder; on Mac inside the app (the installer links it as `reactio` when it can, or prints the command that does it); on Linux **./reactio** in the program folder (install.sh links it into ~/.local/bin). The input is a file (.reactio, .mol, .sdf, .rxn, .smi, .cdxml, .cdx, .json, .docx) or a SMILES in quotes.

Command	What it does
`render <input…> -o out.png`	a picture (.svg, .png, .eps, .tiff); `--scale` for PNG size, `--dpi` for TIFF, `--style ACS` and the other drawing styles
`convert <input…> -o out.cdx`	another format (.reactio, .mol, .sdf, .rxn, .smi, .cdxml, .cdx, .json or a picture); without -o, text formats go to the screen: `--format smi`. A Word file as input gives all its drawings, one under another
`word-clean paper.docx [-o out.docx]`	the Word file with plain vector pictures, without Reactio Draw data
`calc`, `inchi`, `nmr [--nucleus 1H]`, `name`	properties, InChI, predicted shifts or the IUPAC name of every structure in the input
`check <input…>`	valence and drawing problems; exit code 3 when there are errors or warnings
`smiles "name"`	SMILES from a name
`op <command> '{json}' [-i input]`	any command of 6.5

```
reactio render scheme.cdxml -o scheme.png --scale 4
reactio render *.mol -o pictures/ --format svg
reactio convert reaction.cdx -o reaction.rxn
reactio convert paper.docx -o paper.cdxml
reactio calc compounds.sdf --json > properties.json
reactio check scheme.cdxml
```

With several inputs, -o names a folder: one file per input. Only the file or folder after -o is written; answers go to the screen as text, or as JSON with `--json`. `reactio --help` and **Command line.txt** list every option; `reactio --version` gives the version.

6.5 Commands

Command	Arguments	Result
`help`	—	the names of all commands
`version`	—	the name and version of Reactio Draw
`list`	—	every structure on the sheet: index, canonical SMILES, formula, MW, exact mass, role
`properties`	`{index}`	one structure in detail: formula, masses, m/z, elemental analysis, isotope pattern, SMILES
`calc`	`{smiles}`	formula, charge, MW, exact mass, $[M+H]^+$ — nothing is drawn
`inchi`	`{index}` or `{smiles}`	InChI and InChIKey
`nmr`	`{nucleus: "1H"|"13C"|"19F"|"31P", index}` or `{nucleus, smiles}`	predicted shifts with typical errors
`name`	`{index}` or `{smiles}`	IUPAC name; checkedByOpsin tells if it reads back as the same structure (apps only)
`smilesFromName`	`{name}`	SMILES from a common or systematic name (systematic: apps only)
`addSmiles`	`{smiles, x, y}`	draws the structure; @/@@ become wedges
`addName`	`{name, x, y}`	draws the structure for a name
`addReaction`	`{smiles: "A.B>>C"}` or `{reactants: […], products: […], above: […]}`	draws a reaction with its arrow and conditions
`addText`	`{text, x, y, size}`	a caption on the sheet
`addChart`	`{slices: [{label, value, color}], x, y, size, legend, hole}`	a product-ratio chart
`addRadar`	`{axes: [{label, value}], x, y, size, min, max, step, shape, zones}`	a sensitivity screen
`import`	`{format: "smiles"|"mol"|"rxn"|"sdf"|"cdxml"|"cif", text}`	adds the structures of the text
`export`	`{format: "smiles"|"mol"|"sdf"|"rxn"|"cdxml"|"svg"|"json"}`	the drawing in that format
`picture`	`{scale}`	the drawing as a PNG picture (base64); scale 1 = screen size
`set`	`{text, format}`	replaces the whole drawing; the format (smiles, mol, sdf, rxn,

Command	Arguments	Result
		cdxml, json) is found from the text when left out
`image`	`{text, format, output: "svg"|"png", scale}`	a picture of any structure or reaction, without touching the sheet
`selection`	—	what is selected: structure indices, counts, SMILES
`select`	`{index}`, `{indices}`, `{all: true}` or —	selects structures, everything, or nothing
`atoms`	`{index}`	the atoms of a structure, numbered from 1
`highlight`	`{index, atoms, color}` or `{index, clear: true}`	colours atoms and the bonds between them (the whole structure without atoms)
`check`	—	valence errors, unreadable labels and structure checks per structure, each with a level (`err`, `warn`, `hint`)
`saved`, `markSaved`	—	whether there are unsaved changes; mark the drawing as saved
`zoom`	`{value}` (1 = 100 %) or `{fit: true}`	zoom, or fit the drawing in the window
`readOnly`	`{on}`	view only: toolbars and panels hidden, nothing can be changed
`scheme`	—	the reaction: reactants, products, reaction SMILES
`table`	—	the reagent table: equivalents, amounts, masses, volumes, yield
`setAmount`	`{formula|name|id, field, value, limiting}`	fills a cell: field = mass, volume, moles, equivalents, measured, purity, density or molarity
`setRole`	`{formula|name|id, kind}`	the role of a column: reactant, catalyst, solvent or product
`setStyle`	`{style, theme}`	drawing style and colour palette by their names in the program: style ACS, Large, Nature, RSC, Angewandte, Synthesis or Custom; theme Reactio colors, Publication, Pastel…
`prefixStyle`	`{mode: "plain"|"italic"|"sup"}`	t, i, s, n of tBu, iPr…
`clean`	`{what: "reaction"}, {flat: true}` or –	Clean Up Reaction, Clean Up 2D, or Clean Up Structure
`delete`	`{index}`	removes a structure
`undo`, `redo`, `clear`	—	editing

6.6 Good to know

- Structures are numbered from 0 in reading order, as `list` gives them.
- Names need the apps: OPSIN (needs Java) reads names, openclatura (needs Python with openclatura) makes them. In the web version these commands answer with an error; the ~600 built-in names and abbreviations work everywhere.
- In the web version, `nmr` asks the visitor once before downloading the tables (2.2 MB).
- Every command that changes the drawing or the reagent table can be undone with Ctrl+Z, like a change made by hand.

7. Reactio Draw tab in Word

The Windows and Mac apps add a Reactio Draw tab to Word, as reference managers do. Its buttons work on the open document after saving it (a .docx on this computer):

 **Plain pictures**: makes the copy with plain vector pictures next to the document and opens it.

 **Export to CDXML**: saves all structures of the document in one CDXML file next to it, one under another.

 **Open in Reactio Draw**: opens all structures of the document on one sheet in Reactio Draw.

The Drawing group of the tab works on one picture:

 **Edit drawing**: click a Reactio Draw picture once, then press it: the drawing opens in Reactio Draw. On a Mac, Save there (Cmd+S) puts the new picture back in Word, in the same place and at the same scale. A double-click on the picture does the same; if Word does not respond to the double-click, use the Edit drawing button. To give it keys: Tools → Customize Keyboard → Categories: Macros → ReactioDrawEdit. On Windows, double-click the object instead.

 **Update**: puts back drawings saved in Reactio Draw if Word did not take them by itself (Mac). The first time, the Mac asks whether Reactio Draw may control Word: allow it.

 **Mark drawings**: a small blue mark "⌬ Reactio" after every picture with Reactio Draw data. The mark is hidden text: the button turns on Word's display of hidden text, so it shows on screen, and it is never printed. The button stays pressed while the marks are on; press it again to take them away. Plain pictures removes them too.

A drawing put into Word on Windows is a Word object, and one put in on a Mac is a picture: on the other system Edit drawing and the double-click do not open it. There, copy it in Word and paste it into Reactio Draw, or press Open in Reactio Draw to get all structures of the document on one sheet.

A Reactio Draw drawing wider than the text between the page margins, pasted or grown after editing, is scaled down to fit within a few seconds, keeping its proportions (Windows and Mac).

The tab is the template "Reactio Draw.dotm" in Word's Startup folder. Word may ask once whether to enable its macros: the buttons work only with them on. On a Mac, Word starts other programs only through a script in its own scripts folder, so the installer puts a small helper script there too.

Install: on Windows, ReactioDraw.exe adds the tab when it starts (restart Word once); on a Mac, Install Reactio Draw.command adds it when Word is on the Mac, and "Install Word tab.command" in the Reactio Draw folder adds it again (with --remove it takes it away).